LOOKUP, NOT RELAY: HOW LANGUAGE MODELS USE WHAT THEY HAVE NOT WRITTEN

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Paper under double-blind review

ABSTRACT

Language models often use information they have not written down, such as a planned rhyme, an intermediate result or a secret choice. It can reach the position that uses it in three ways: written into the text and read back (emission), looked up at a fixed position of the context, such as where it originates (lookup), or passed forward in the hidden states of the text the model generates, without being written (relay). Relay would undermine chain-of-thought monitoring, because a monitor reading the generated text would not see the information. We introduce *route accounting*, which edits hidden states at the source and measures how much of the effect travels by each path. On induction, where information is known to reach its target through the hidden state of an intermediate position, it finds that path as the largest in all nine models tested. On four tasks across 15 checkpoints of Qwen3, Qwen3.5 and Gemma 3 (1B to 32B parameters), with thinking disabled and short generated spans, models instead look up information. Rhyme plans are read from the rhyme word or from a copy stored at the line's final token, and relay accounts for at most 1% of the effect. Patching every position after a computed value's source barely moves the answer, while written values account for at least 83% of the effect. Three stress tests bound this pattern: beyond the attention window the model hands the plan to the stored copy, which becomes necessary for about half of the effect, and the generated text accounts for 9 to 13%; entry through the recurrent layers of a hybrid produces 2 to 4%; and at most 2% of a secret choice's effect passes through the hidden states of the intervening sentence. The stored copy is the one path that grows with scale. In these settings, unwritten information is kept at a few predictable positions of the context rather than relayed through the hidden states of the generated text: invisible to a monitor reading the text, but within reach of one that probes hidden states.

1 INTRODUCTION

Language models routinely act on information they have not written down: a model completing a couplet selects its rhyme before writing the line that ends in it (Lindsey et al., 2025; Hanna & Ameisen, 2026a), and a model asked for a calculation or a secret choice must hold or reconstruct that content across the tokens it writes in between. How such information reaches the position that uses it matters for chain-of-thought monitoring, which assumes that decision-relevant reasoning passes through, or can be recovered from, the text the model writes (Korbak et al., 2025; Baker et al., 2025). The central risk to that assumption is content the model uses but never states (Pfau et al., 2024; Roger & Greenblatt, 2023; Turpin et al., 2023; Chen et al., 2025b).

In a transformer, information at one position can reach a later position in three ways (Figure 1). It can be **emitted**: it changes the tokens the model writes, which are read back as input. It can be **looked up**: read from a fixed position of the context, where it originates or in visible text from which it can be recomputed. Or it can be **relayed**: the hidden states of the generated positions pick it up and pass it on, although the words written there do not express it. Emitted information is in the text and looked-up information stays at a fixed position of the context, where it can be read or probed; relay is the path by which a model could hold content across a chain of thought that never states it. Lookup turns out to have a second site: a **stored copy**, written once into a structural position of the prompt next to the source. Prior work decodes upcoming tokens from hidden states (Pal et al., 2023) and locates where plans are formed or can be steered (Lindsey et al., 2025; Hanna & Ameisen, 2026a;

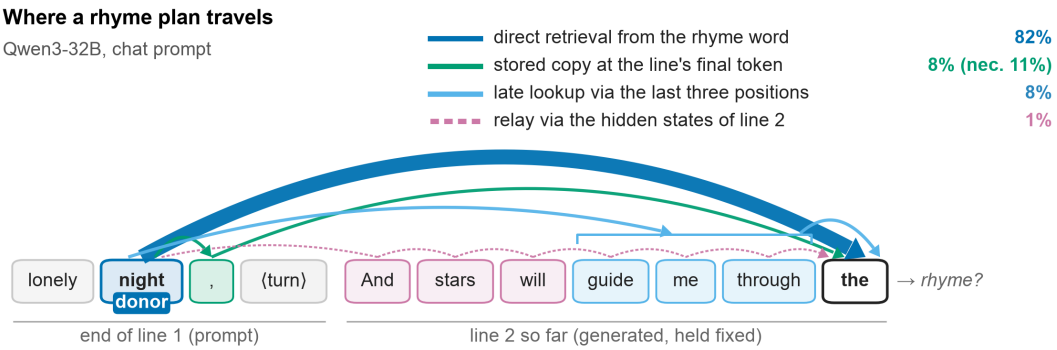


Figure 1: Route accounting on a rhyme plan. The paths by which a plan formed at the rhyme word of line 1 (where the donor state is written) reaches the position that predicts the rhyme, with the line-2 words held fixed. Widths are each path's measured share of persistence in Qwen3-32B with the chat prompt: most of the plan is read directly from the rhyme word, and relay through the hidden states of line 2 accounts for 1%. Appendix Figure A1 gives, for every setting and model, the looked-up share and the share of relay or of an unsplit indirect path.

Maar et al., 2026), but does not measure how much of a plan's effect takes each path to where it is used.

We introduce **route accounting** (Section 2). A donor's hidden states are written at the source positions, and the effect is measured where the information is used. Fixing the generated text separates emission from persistence. Clamping the states of chosen groups of positions, at every layer, separates direct retrieval of the source from indirect paths through intermediate positions, which we split by position into the stored copy, the late lookup just before the target, and relay through the rest of the generated text. Direct retrieval, the stored copy and the late lookup all read the information from a fixed prompt position, so they are the three forms of lookup; relay is the only path that holds it across the generated text. On induction (Olsson et al., 2022), where the known mechanism passes information through the hidden state at the position of a copied token, route accounting finds that indirect path as the largest path in all nine models tested (Section 2.1).

Applied to four tasks in 15 checkpoints from three model families, route accounting finds one principle, within the settings tested (short generated spans, thinking disabled): **lookup, not relay**. Models read unwritten information near the point of use from a fixed position of the prompt, the source or a copy stored next to it, rather than relay it through the hidden states of the text they generate (Table 1; Section 3). Rhyme plans are read mainly from the rhyme word (68 to 95% of persistence in all but one configuration), and relay accounts for at most 1% in all 21 runs. Donor states at every position after a computed value's source move the answer by at most 4.9% (variable chains) and 6.4% (sums) of the source effect, while written intermediate values are followed: emission accounts for at least 83% of the effect.

Three stress tests with committed decision rules make relay useful or the content invisible (Section 4). Beyond Gemma 3's attention window, Gemma 3 27B hands the plan from the rhyme word to the stored copy, and the generated text accounts for 9 to 13%, not split into late lookup and relay. In a recurrent hybrid, entry through the recurrent layers produces 2 to 4% of persistence. A secret choice is stored at a few low-content prompt tokens, and at most 2% of its effect passes through the hidden states of the intervening sentence. The one path that grows is the stored copy: it appears at 14B and above in Qwen3 and at 12B to 27B in Gemma 3, is present in base checkpoints, and reconciles published accounts of where the rhyme plan lives.

Contributions. (1) Route accounting, a position-level decomposition of intervention effects into emission, lookup and relay, validated on a positive control. (2) The principle that unwritten information is looked up, not relayed, across tasks, families, sizes, prompt formats, checkpoints and intervention types. (3) Three stress tests with committed decision rules that bound it. (4) A stored copy that emerges with scale, reconciling published accounts of rhyme planning.

Table 1: Where unwritten information travels: the share of the effect by path, as a range over the models tested, of persistence unless stated. *Looked up*: the target reads the source, a copy stored at a structural token next to it, or the late lookup. *Relay, or an unsplit indirect path*: relay where it was separated (rhyme plans); otherwise the named group of intermediate positions. *Visible* content can be recovered by a reader of the text; *hidden* content cannot. Dashes mark cells the design rules out. A share above 100% means that the other paths, or their interaction, are negative. The sections and appendix give the full results with 95% intervals.

Content (observability)	Emission	Looked up: direct retrieval, stored copy or late lookup	Relay, or an unsplit indirect path
Rhyme plan (visible)	13–25% of the total effect	rhyme word 68–95% of persistence (exception: 19% in Gemma 3 27B, plain, where the stored copy is 37% by sufficiency and 74% by necessity); stored copy otherwise 0–16% by sufficiency; late lookup 1–13%	relay $\leq$1%
Rhyme plan after 500–2,000 filler tokens (Gemma 3 27B)	–	rhyme word 40–64%; stored copy necessary for 25–49%	through the generated text, late lookup and relay together: 9–13%
Computed value, unwritten (visible)	–	source edits effective; downstream patch effect near zero	every later position together: $\leq$4.9% of the source effect (chains), $\leq$6.4% (sums)
Computed value, written (visible)	83–101%	bypass to the source $-$1–17%	–
Secret choice (hidden)	no measurable effect (the sentence never names it)	pick-specific information stored after the list: 0–40% of the text-swap reference, read directly at the reveal (Gemma 4 31B stores 82–91%; its path split was not run); the rest not reproduced by the post-list patch	through the sentence: $\leq$2% of the text swap ($\leq$16% of the stored part); sentence positions $\leq$8% (Qwen3.5 hybrids)
Induction (positive control)	–	direct retrieval $\leq$3% (10–31% in Qwen3-32B and the Qwen3.5 hybrids)	through the copied token's position (prompt): 53–137%

2 Route accounting

Setting. A model processes a prompt and generates text. An intervention changes the information at a set of **source** positions S, and its effect is measured at a later **target** position t where that information is used. The positions strictly between them, M, include the end of the prompt and the tokens generated so far. Every intervention is an **all-layer state edit**: the residual stream at every layer of each source position is replaced by the states of a *donor* prompt that differs only in the content under study (a first line with a different rhyme, a list in a different order), while the prompt text is unchanged. The outcome is a log-odds contrast at the target,

$$R \;=\; \log \sum_{v \in \mathcal{V}_{\text{donor}}} p(v \mid \cdot) \;-\; \log \sum_{v \in \mathcal{V}_{\text{orig}}} p(v \mid \cdot),$$

where $\mathcal{V}_{\text{donor}}$ and $\mathcal{V}_{\text{orig}}$ are the tokens consistent with the donor's and the original content, for example the single-token perfect rhymes of each rhyme word. Effects are changes in R relative to the unedited run; positive values move the target toward the donor.

Emission and persistence. The edit can change what the model writes, and the written tokens are read back. Let g_0 and g_1 be the texts generated with the edit off and on, and $R(i, g)$ the outcome with the edit in state $i \in \{0, 1\}$ and the text fixed to g. Then

$$\underbrace{R(1, g_1) - R(0, g_0)}_{\text{total}} \;=\; \underbrace{R(0, g_1) - R(0, g_0)}_{\text{emission}} \;+\; \underbrace{R(1, g_1) - R(0, g_1)}_{\text{persistence}}.$$

Emission replays the edited words with the edit off. **Persistence** is what the edit does with the text held fixed. We also report persistence with the original text, $R(1, g_0) - R(0, g_0)$, the other order of the decomposition, where the two differ (Appendix Section C gives the single-token form).

Paths within persistence. The target can read the edited source itself (**direct retrieval**), or the hidden states of intermediate positions can read it and pass it on (an **indirect path**). We separate the two by clamping states. Let h^0 and h^1 be the per-layer states of the clean and the edited run, both with the text fixed, and write $R[\,S \leftarrow a,\ G \leftarrow b\,]$ for the outcome when the source receives the states of run a and a group $G \subseteq M$ those of run b, at every layer, with all other positions recomputed. With R_0 the clean outcome,

$P = R[S \leftarrow 1] - R_0$	persistence,
$D = R[S \leftarrow 1,\ M \leftarrow 0] - R_0$	direct retrieval (every intermediate position clean),
$T = R[M \leftarrow 1] - R_0$	indirect paths (source clean, intermediates as in the edited run),
$\Delta = P - D - T$	interaction (additivity gap).

Because the intermediate states are overwritten at every layer, D reaches the target only through its own attention to the source, and T only through intermediate positions. For a group G we report its **interventional sufficiency**, $\mathrm{Suf}(G) = R[G \leftarrow 1] - R_0$, the effect of the group alone, and its **interventional necessity**, $\mathrm{Nec}(G) = P - \big(R[S \leftarrow 1,\ G \leftarrow 0] - R_0\big)$, the loss of persistence when the group alone is reset to its clean states. When paths combine additively, the two agree and shares partition the effect. Where the interaction is large (at most 33% of persistence, in Gemma 3 27B couplets), shares are not exclusive, and we report both measures and their average, the two-player Shapley value (Shapley, 1953). Shares are divided by the persistence of the same run unless another denominator is named, and intervals are item-bootstrap 95% intervals (Efron & Tibshirani, 1994).

Path names. We split M into three groups by position, not by any hypothesis about their content, and name the indirect path through each. The **stored copy** passes through the prompt positions right after the source line (in couplets, the token that ends line 1 and the chat template). The **late lookup** passes through the three positions just before the target; it is the late retrieval of Hanna & Ameisen (2026a) read in two steps. **Relay** passes through the remaining positions of the generated text; it is the path monitoring is concerned with. Because the text is held fixed, every indirect path, relay included, passes information only in hidden states; information in the written words themselves is emission. Information is **looked up** when it arrives by direct retrieval, the stored copy or the late lookup. Where a group was not split further, we name it and call its effect the indirect path through it.

2.1 Positive control: induction

A decomposition that never reports an indirect path could be blind to one. In a repeated sequence $\ldots A\,B \ldots A$, a previous-token head writes "preceded by A" into the position of B, and an induction head at the second A finds that position and copies B (Olsson et al., 2022). Information about the first A therefore reaches the prediction through the hidden state of an intermediate position, while the token B itself is unchanged. We edit the first A in 100 random sequences and measure $\log p(B)$ at the second A. The committed criterion (an indirect path above 80% of persistence, direct retrieval near zero) holds cleanly in Qwen3 1.7B to 14B (indirect path 108 to 137%, the excess an interaction; direct retrieval 1 to 3%). It holds partly in Gemma 3 12B and 27B, where conflicting edited positions lower the all-position cell to 53 to 55%, and in Qwen3-32B and the Qwen3.5 hybrids, which also read the source directly (10 to 31%). In every model the indirect path is the largest path (Appendix Section E). The key position lies right after the source, so this path resembles a stored copy more than relay; but with the generated text fixed, generated tokens are ordinary context positions, and the clamp is the same operation at any position. The lookup-dominated results that follow are therefore informative negatives, not a blind spot of the method.

Models and protocol. We study Qwen3 1.7B to 32B (Yang et al., 2025a), Gemma 3 1B to 27B (Gemma Team, 2025), the recurrent hybrids Qwen3.5 4B, 9B and 27B (Qwen Team, 2026), and Gemma 4 31B (Gemma Team, 2026) for the hidden choice, as instruction-tuned checkpoints with base checkpoints as checks and thinking disabled. After the initial couplet split, the design and decision rule of every experiment were committed to the repository before its first result, and later additions are labelled post hoc (Appendix Section D; materials in Appendix Section B).

3 UNWRITTEN INFORMATION IS LOOKED UP

3.1 RHYME PLANS

Rhyming couplets are the flagship case of latent planning (Lindsey et al., 2025; Hanna & Ameisen, 2026a). We use the couplet prompts of Hanna & Ameisen (2026a) in their chat format and repeat every run with the plain prompt of Ma & Rui (2026), which has no chat template. *Chat* and *plain* name prompt formats, not checkpoints: instruction-tuned models are run with both, and base checkpoints, which have no chat template, with the plain prompt only (100 couplets per run; Appendix Section F). The source is the last word of line 1, the donor a couplet from a different rhyme group, and the target the position that predicts the first token of line 2's last word, with the line-2 prefix fixed. The edit moves line 2 off its rhyme (at most 2% of second lines still rhyme with the original, against 83 to 100% before the edit) and onto the donor's rhyme in 27 to 95% of couplets. The emitted line-2 words account for 13 to 25% of the total effect; the rest is persistence.

Direct retrieval dominates, and relay is at most 1%. Figure 2 decomposes persistence in all 21 runs. Direct retrieval of the rhyme word accounts for 76 to 95% of persistence in every Qwen3 configuration (1.7B to 32B, chat and plain, instruct and base) and 80 to 95% in Gemma 3 1B to 12B. The late lookup accounts for 1 to 13%. Relay through the rest of the generated line is at most 1% of persistence in every run, by sufficiency in all 21 and by necessity in the 17 where it was measured. Before running the larger models we committed two criteria for the claim that relay grows with scale: the indirect share must rise with log parameter count across Qwen3 1.7B to 32B, and the rise must come from relay. The first holds (6.2 percentage points per decade, 95% interval [2.8, 9.6]); the second fails (relay is -0.06 to $+0.16$, against persistence of 12 to 26). The indirect share grows as the late lookup up to 8B and as the stored copy from 14B. Controls support the split (Appendix Section F): same-rhyme donors leave the rhyme almost unchanged, the rhyme word matters two to three times more than any other position of line 1, and the published feature steering of Hanna & Ameisen (2026a) gives the same split as the state edit (direct retrieval 90, 88 and 84% at 4B, 8B and 14B, against 91, 90 and 82%).

3.2 THE STORED COPY

Gemma 3 27B is the exception to the dominance of direct retrieval. Direct retrieval falls to 68% of persistence with the chat prompt and 19% with the plain prompt, and two components appear: a large interaction and a copy stored at the line's final token.

It sits on the line's final token. Resetting the tokens between the rhyme word and line 2 one at a time locates the copy. In all five runs analysed, the line's final token accounts for 90 to 100% of these tokens' summed necessity, and the chat-template tokens hold almost nothing (Appendix Figure A2). With the chat prompt this token is the comma that ends line 1; with the plain prompt, the newline. The site is the last token of the line, whatever that token is; in the prompt of Lindsey et al. (2025) it is the newline, where they observed Claude's rhyme plan being formed.

It emerges with scale, before instruction tuning. In Qwen3 the copy is absent up to 8B (necessity at most 0.9% of persistence) and appears at 14B (8% with the chat prompt and 16% with the plain prompt, by necessity), remaining at 32B (11% and 13%). In Gemma 3 it is small at 12B (1 to 3%) and present at 27B: 11% by necessity with the chat prompt and, with the plain prompt, 37% by sufficiency and 74% by necessity, more than is read from the rhyme word (19%), while relay stays at 1%. The base checkpoints we tested have the copy wherever the matching instruct model does (Qwen3-14B-Base 16% against 13%; Gemma 3 12B PT 5% against 2%), so it most likely arises in pretraining. Same-rhyme donors remove it, so it encodes the rhyme.

Where paths interact, sufficiency understates it. In Gemma 3 27B with the chat prompt, the copy alone moves the rhyme by only $+0.68$, yet resetting it removes $+4.87$ [3.18, 6.73] of a persistence of $+43.1$; the interaction is 29% of persistence (Qwen3 at most 5%). The target combines the rhyme word and the copy, and reads conflicting evidence when only one of them is edited: the copy is a redundant second lookup site.

Reconciling published accounts. The heads near the end of line 2 that read the rhyme word (Hanna & Ameisen, 2026a) are direct retrieval with the late lookup; the plan found and steered at the end of line 1 (Lindsey et al., 2025; Maar et al., 2026) is the stored copy of larger models; the hand-off to the newline in Gemma 3 27B only (Ma & Rui, 2026) matches our plain-format result; and no copy

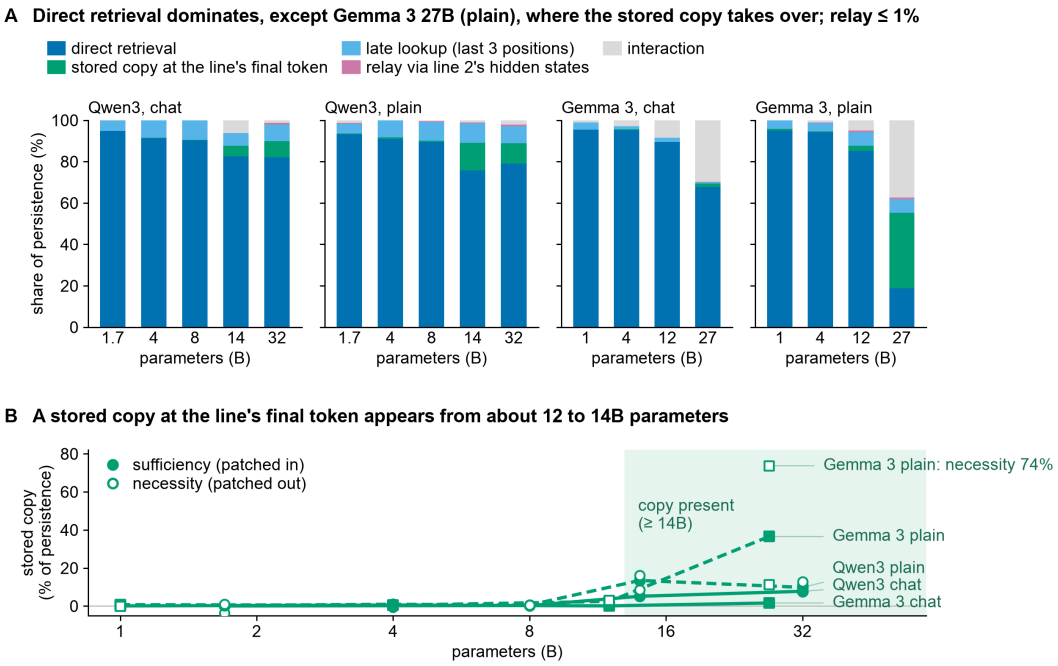


Figure 2: Couplet paths across scale, family and prompt format (100 couplets per run; instruction-tuned checkpoints; Appendix Table A4 adds the base checkpoints). **Top:** composition of persistence from the sufficiency cells: direct retrieval of the rhyme word (blue), the stored copy at the line's final token (green), the late lookup (light blue), relay through the rest of line 2 (purple; at most 1%, so barely visible) and interaction (grey), one bar per model. **Bottom:** the stored copy's share of persistence against model size (log scale), by sufficiency (filled) and necessity (hollow). It is absent up to 8B and present from 14B in Qwen3 and at 12B to 27B in Gemma 3.

appears below 8B, consistent with the inert newline that Jacopin (2026) finds at 0.6B to 2.6B. Each account is correct for the configuration it studies (Appendix Section F).

Why the stored copy is a lookup, not relay. It passes through an intermediate position, but it is a fixed structural token next to the source. The copy is written there once, read directly however much text intervenes, and does not move with generation. It never travels through the hidden states of the generated text, so a monitor probing hidden states can find it at a known prompt position.

3.3 Computed values

Computed information is not present at any single token, which is where an indirect path would seem most useful (Appendix Section G). In two-digit sums (Qwen3 1.7B to 32B, 104 items, a filler of 0 to 32 tokens before the answer), donor states at every position from the question mark to the answer move the answer by at most 6.4% of the effect of donor operand digits. Variable chains (`a = 23, b = a + 4, ..., What is e?`) make recomputation at the answer costly and an indirect path, passing the running value from statement to statement, the natural one, so we designed them as a second positive-control candidate for an indirect path. They are not one. Donor states at every position after the starting value together, a test that would detect a running value stored redundantly, move the answer by at most 4.9% of the effect of editing the starting value (32B, one update), and their interval includes zero at every size with five updates, where a running value would help most. This is consistent with the answer re-reading its visible sources; recomputation itself is not measured.

3.4 Written values

When asked to compute each variable in order, the model writes the donor's values in 86 to 100% of items, and the answer follows the text. With the chain cut before its last value, emission accounts for 83 to 101% of the effect at 1.7B to 32B, and the bypass that re-reads the source stays below 17% (the 101%, at 32B with five updates, reflects a slightly negative bypass). For a binary mediator such as the article *a* or *an* before a planned noun, emission factorizes exactly into the change in

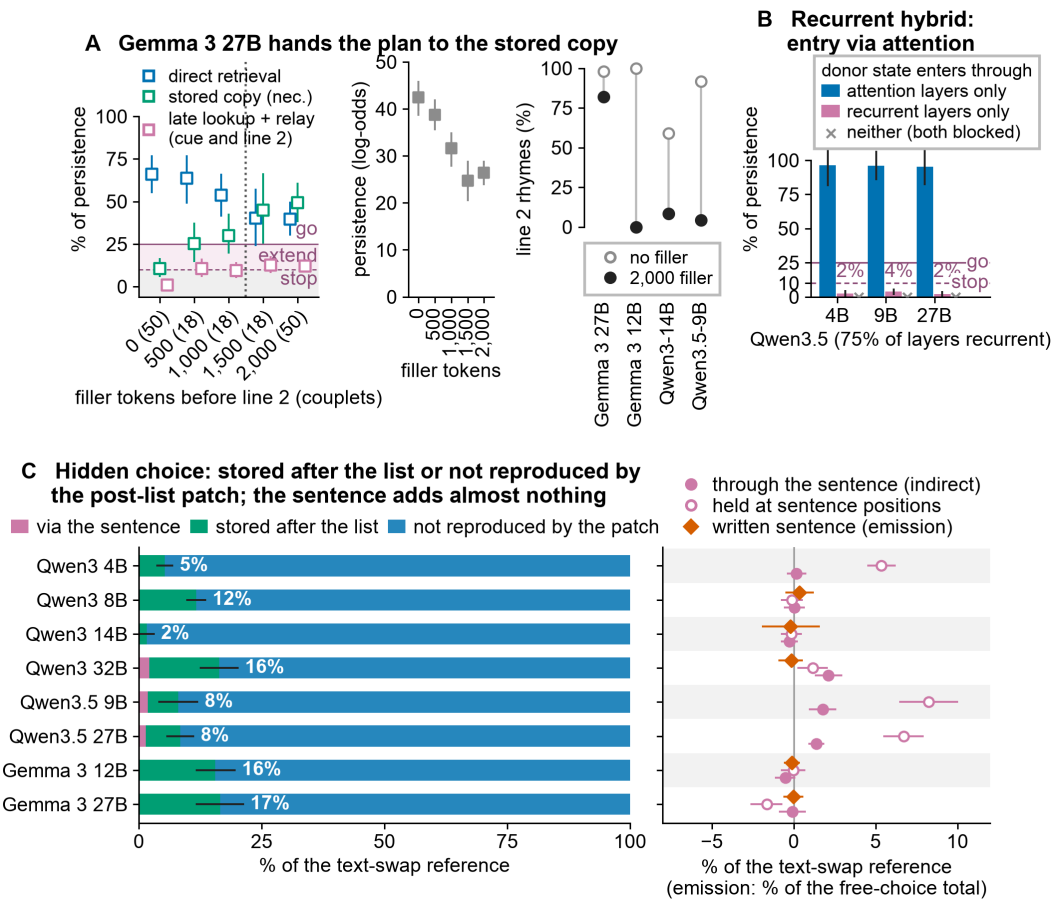


Figure 3: In settings built to favour relay, models still look up information. **A:** Gemma 3 27B with filler between line 1 and line 2 (couplets per distance in brackets; the dotted line marks where line 1 leaves the 1,024-token window). Direct retrieval falls while the stored copy becomes necessary for about half of persistence; the generated text (the cue and line 2, not split into late lookup and relay) is 9 to 13%, between the committed stop and go lines. Side panels: absolute persistence, and for four unedited models the share of couplets whose line 2 still rhymes, with no filler (open) and after 2,000 filler tokens (filled). **B:** Qwen3.5 hybrids (three of every four layers recurrent), with the donor state allowed to enter only one layer type at the source. Entry through the attention layers accounts for nearly all of persistence; entry through the recurrent layers produces 2 to 4%, below the committed stop line (10%). **C:** hidden choice, as shares of the text-swap reference: not reproduced by the post-list patch, stored after the list, and passed through the sentence. The right subpanel shares the bars' rows and magnifies, with 95% intervals, the indirect path through the sentence, the pick held at the sentence positions, and emission through a sentence the model writes (share of that run's total).

what is written times the written token's **leverage**, how much it matters to what follows (Appendix Section G). For the a/an planning features of Hanna & Ameisen (2026a) in Qwen3-4B, 92 to 94% of their feature-specific effect on the planned noun travels through the generated article, the bypass is at the level of random features, and the effect through the article scales with leverage.

4 Settings built to favour relay

Relay is not required when every layer can read every earlier position and the content is visible. We therefore run three stress tests that remove one of these conditions, each with a decision rule committed before its first result (Figure 3; Appendix Section D, Section I and Section J).

4.1 BEYOND THE ATTENTION WINDOW

Gemma 3 alternates five local sliding-window layers (Beltagy et al., 2020) with a 1,024-token window with one global layer (Gemma Team, 2025), so once line 1 lies more than 1,024 tokens before the target, five of six layers cannot read it. We insert up to 2,000 tokens of neutral filler before line 2 and split the indirect path into the stored copy, the filler and the generated text (the cue and line 2). The committed decision number was the share of persistence through the generated text: "go" at 25%, "stop" below 10%, and an extension to 50 couplets in between; a rise counts only if the absolute indirect effect also grows. At 2,000 tokens Gemma 3 27B still rhymes in 82% of couplets, and it hands the plan from the rhyme word to the stored copy at the line's final token. Direct retrieval falls from 66% to 40% of persistence ($+28.0$ to $+10.5$), while the copy's necessity rises from 11% to 49% ($+4.6$ to $+13.0$ [10.0, 16.1]) although persistence falls from $+42.4$ to $+26.4$; the copy alone moves the rhyme by $+0.1$ without filler and by $+9.7$ with it. The rise is absolute, so it passes the committed rule; the filler passes on nothing. The generated text rises from 1% to about 11% by sufficiency as soon as filler is present (12% at 2,000 tokens, $+3.2$ [2.2, 4.3]) and does not grow further. It includes the last three positions, so it mixes relay with the late lookup, and it falls between "stop" and "go": how much of it is relay remains unresolved. The committed window prediction is only partly supported: the largest step is where line 1 leaves the window, but the shift begins at 500 tokens. Models without a strong stored copy (Gemma 3 12B, and Qwen3.5-9B and Qwen3-14B, whose attention has no window) lose the plan instead, and no indirect path takes over.

4.2 RECURRENT MEMORY

Qwen3.5 (Qwen Team, 2026) interleaves three Gated DeltaNet layers (Yang et al., 2025b) with each attention layer. In a recurrent layer, information from the source can reach a later position only through a fixed-size state updated position by position, invisible in the text; the attention layers can still look it up. The test asks which path the model uses when both are available. In the *attention-only* cell, the donor state enters at the source but the input of every recurrent layer there is restored to its clean value; the *recurrent-only* cell is the converse. The committed rule, decided on the largest model, was "go" if recurrent-only is at least 25% of persistence and "stop" below 10%. Recurrent-only is 3% and 4% of persistence at 4B and 9B, and 2% at 27B ($+0.41$ [0.18, 0.64] of $+18.8$); attention-only is 95 to 97%, and blocking both cancels the donor on every couplet. Decision: stop. The design controls which layer type receives the donor at the source, not every later path through each type, so it bounds donor entry through recurrent memory at 2 to 4% of persistence.

4.3 HIDDEN CONTENT

A choice that no reader can infer from the text is the case monitoring cares about most. The prompt reads: "Secretly choose one fruit from this list: {4 fruits}. Do not write your choice yet. First write one sentence about the weather. Then write the fruit you chose." The greedy pick depends on the model's preferences and the list order, so no reader can compute it. The donor is the list in another order with a different pick; the reference is a *text swap* that writes that order into the prompt. In Qwen3-8B and Gemma 3 12B and 27B, donor states at the positions after the list move the reveal by 12 to 16% of the text swap, all of it read directly; the part that passes through the sentence is zero within its interval, and the rest of the pick is presumably re-derived at the reveal from the visible list. In Qwen3-32B and the Qwen3.5 hybrids the indirect path through the sentence is small but nonzero: at most 2% of the text swap, but one eighth to one fifth of the stored part. A sentence the model writes itself never names an item and has no measurable effect on the reveal (emission -0.01 to $+0.02$).

The stored signal is pick-specific and sits at predictable tokens. In an out-of-list control, the donor's pick does not occur in the prompt, yet post-list donor states raise it more than the donor's other items. This *pick specificity*, added post hoc for fruits, was then committed as the primary measure of a replication with animals, which confirms it in both families; a donor with the same pick leaves the preference unchanged. The stored share is larger in Gemma and grows with generation (Gemma 3 15 to 40%, Gemma 4 31B 82 to 91% of the text swap's pick specificity), and it shrinks with scale in Qwen3 (0 to 7% at 14B and 32B). Per-token necessity places it on two or three low-content tokens: the periods that close the list or the instruction, and one function word or possessive token of the instruction (in Gemma 3 27B, 77% on *the* in "about the weather"). When the instruction is reworded, the site moves with the wording at 4B, 12B and 27B (Appendix Figure A2). These controls do not distinguish a stored pick from a stored copy of the list, so we claim only that pick-specific information absent from the text is stored after the list and read at the reveal. As for rhymes, the copy is stored once, at a segment-closing or predictable token, and read at the point of use.

5 Related work

Planning. Hidden states carry information about tokens several positions ahead (Pal et al., 2023; Men et al., 2024), much of it a by-product of current-token computation (Wu et al., 2024), and attribution graphs (Ameisen et al., 2025) expose planning in rhyming (Lindsey et al., 2025) and article choice (Hanna & Ameisen, 2026a). Hanna & Ameisen (2026a) show that such planning emerges with scale in Qwen3 and identify the circuit that retrieves the rhyme plan near the end of line 2; other studies locate or steer the plan at the line boundary (Maar et al., 2026; Ma & Rui, 2026), which Jacopin (2026) finds inert at 0.6B to 2.6B. These works ask whether and where a plan exists. We ask which path it takes, and show that the published accounts measure different lookup sites.

Causal mediation and patching. Route accounting builds on activation patching and causal mediation (Vig et al., 2020; Meng et al., 2022; Geiger et al., 2025; Chan et al., 2022; Zhang & Nanda, 2024; Heimersheim & Nanda, 2024) and is closest to path patching (Wang et al., 2023a; Goldowsky-Dill et al., 2023) and attention knockout (Geva et al., 2023). It defines paths over positions rather than components, separates emission through generated text from persistence, and reports sufficiency, necessity and interaction side by side, in the spirit of evaluating mechanisms against explicit mediators (Mueller et al., 2026).

Structural tokens. Label words (Wang et al., 2023b), separators (Chen et al., 2025a) and registers (Darcet et al., 2024) aggregate information, while attention sinks absorb surplus attention (Xiao et al., 2024); the stored copies we find are causal instances that grow with scale and distance.

Chain-of-thought faithfulness and monitoring. Written reasoning can be unfaithful to the computation behind an answer (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025b); models can learn to compute with filler tokens (Pfau et al., 2024) and could in principle encode reasoning steganographically (Roger & Greenblatt, 2023). Monitoring is valuable but fragile (Korbak et al., 2025; Baker et al., 2025), and more robust when reasoning is necessary for the task (Emmons et al., 2025). These studies measure behaviour; route accounting measures internal paths, and our hidden choice, unlike secrets installed by training (Cywiński et al., 2025), arises without it.

6 Discussion

Implications for monitoring. In the settings tested, what a model writes, it uses, in proportion to the leverage of the written token, and what it does not write, it looks up from a few prompt positions: the sources and copies stored at structural tokens next to them. If this extends to other tasks, a monitor of internal state could focus on those positions and track whether stored copies keep growing with scale. The advice is weakest where the indirect path through generated text is largest: 9 to 13% of persistence after filler, not yet split into late lookup and relay, and one eighth to one fifth of the small stored hidden part in Qwen3-32B and the Qwen3.5 hybrids.

Why look up rather than relay? Attention makes any earlier position readable at every layer, so a later position can compute what it needs when it needs it. But at a distance the model relies on a stored copy or loses the plan, and even where three quarters of the layers can pass a plan forward only through a recurrent state, entry through the attention layers accounts for nearly all of the effect. Relay appears disfavoured when lookup is available, not merely unneeded.

What would overturn the principle. Much longer generated text, thinking enabled, tasks that cannot be recomputed from the prompt, or training that rewards concealment (Baker et al., 2025) could break it, and pause tokens (Goyal et al., 2024) or continuous latent reasoning (Hao et al., 2025) are designed to do what we rarely see. Route accounting applies to each case.

6.1 Limitations

The generated spans are short, thinking is disabled, and the tasks are constructed; the principle is established only for them. All-layer state edits are strong and can take a model off its natural distribution, which the feature-steering replication and the same-rhyme nulls address only in part. The hidden-choice controls do not distinguish a stored pick from a stored copy of the list. The models span 1B to 32B parameters in three families, with Gemma 4 31B for the hidden choice only. Recomputation at the answer is inferred, not measured. Five committed checks were not run, and post hoc elements are labelled (Appendix Section D).

AI USE STATEMENT

We used AI coding agents and assistants (Claude, Codex and Cursor) extensively in this work. They wrote and ran experiment code, managed jobs on a remote workstation, aggregated results, generated the figures from the committed result files, and drafted and edited manuscript text. Research questions, experimental designs and decision rules were discussed with the agents. After the initial couplet split, every design and decision rule was committed to version control before the corresponding result was computed. The authors reviewed the code, the committed results and every claim in the manuscript. Numbers in the text were checked against the machine-readable result files, and citations were checked against their sources. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies how openly released language models use information they have not written. Its aim is to support the monitoring of such models. It involves no human subjects, personal data or harmful content. The hidden-choice task elicits harmless secret choices (fruits, animals). The results locate where content that the text does not reveal is stored, which is information useful to monitors. We do not see a plausible route by which it helps a model conceal content. The limitations section states where the findings may not transfer, notably to long reasoning with thinking enabled. The discussion notes that training which rewards concealment is untested.

REPRODUCIBILITY STATEMENT

All experiments use publicly released checkpoints (Appendix Section B) and public prompts released by Hanna & Ameisen (2026a; Hanna & Ameisen, 2026b) (used in their chat format and in the plain template of Ma & Rui (2026)) or generated with fixed seeds. Upon acceptance we will release the code and result files on GitHub. That release will contain, for every experiment, the code, the per-item result rows and summaries with bootstrap intervals, and the scripts that generate every figure and table from those files. After the initial couplet split, it will also contain each design and decision rule as committed before the corresponding result. The log of committed decision rules (Appendix Section D) lists each frozen design, its decision rule and the outcome. Dated deviations are recorded with each experiment, and the main ones are listed there. Batched edits and generation were checked against unbatched reference implementations (Appendix Section B). Compute is described in Appendix Section K.

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A All settings at a glance

B Models and materials

Models. Qwen3 1.7B, 4B, 8B, 14B and 32B (`Qwen/Qwen3-*`), and Qwen3 8B and 14B base (`Qwen/Qwen3-*-Base`). Gemma 3 1B, 4B, 12B and 27B instruction-tuned (`google/gemma-3-*-it`), and Gemma 3 12B pretrained (`google/gemma-3-12b-pt`). Qwen3.5 4B, 9B and 27B (`Qwen/Qwen3.5-*`). Gemma 4 31B (`google/gemma-4-31B-it`) for the hidden choice (Gemma Team, 2026). Qwen3 0.6B, 1.7B and 4B for the a/an analysis, with the per-layer transcoders (Dunefsky et al., 2024) used by Hanna & Ameisen (2026a) and released with circuit-tracer (Hanna et al., 2025). All models run in bfloat16 on Apple silicon (PyTorch MPS), with full recomputation and no key-value cache across edited runs. Where effects are small, the output projection is recomputed in float32, because bfloat16 logits resolve only about 0.125 nats.

Couplet prompts. The chat format of Hanna & Ameisen (2026a). The user turn is "Write only the next line of this rhyming couplet: {line 1}", preceded by `/no_think` for Qwen3. The plain format of Ma & Rui (2026) is `A rhyming couplet:\n{line 1}\n`, with no template; line 2 is the continuation up to the next newline. Chat and plain are prompt formats, not checkpoint types: instruction-tuned checkpoints are run with both, base checkpoints with the plain format only. Rhymes use the perfect-rhyme sets of the Datamuse API, following the released code of Hanna & Ameisen (2026a). The outcome uses the single-token words of each set.

Item construction. Qwen3 1.7B to 14B use the released 100-couplet steering subsets and donor pairings of Hanna & Ameisen (2026a). Other models use their rule: the first 100 first lines, in their

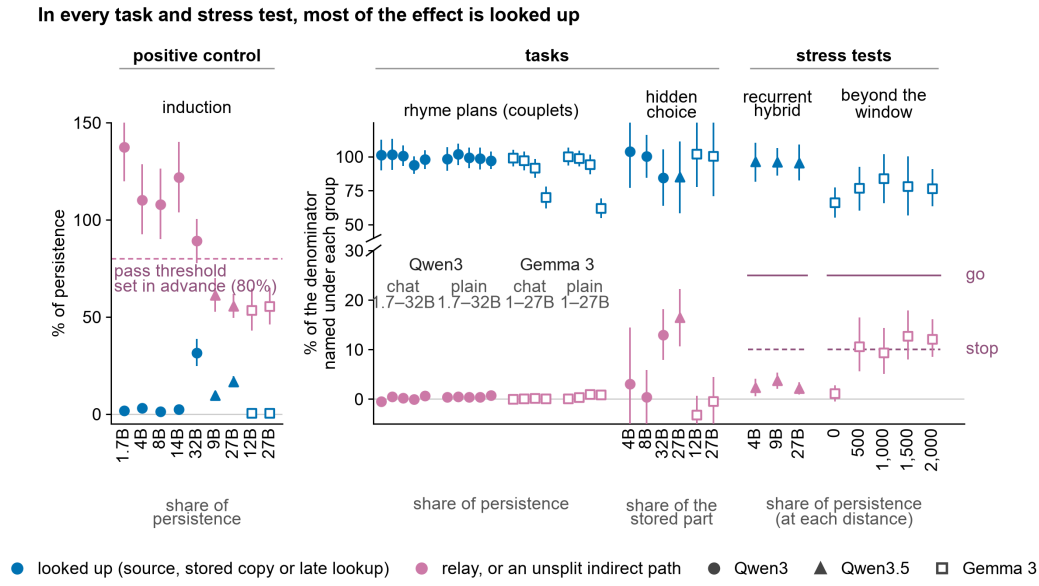


Figure A1: In every task and stress test, most of the effect is looked up. For every setting and model: the share of the effect that is looked up (blue) and the share that is relayed, or takes an indirect path not split further (purple), with 95% intervals. The positive control has its own axis (0 to 150%), with the pass threshold set before the experiment (an indirect path above 80% of persistence). The other settings share a broken axis: looked-up shares on the upper segment, purple shares on the lower segment (to 30%), with the committed stop (10%) and go (25%) lines of the stress tests. The group of intermediate positions and the denominator are given under each group; for the recurrent hybrid, blue and purple are the shares entering through the attention and the recurrent layers; Qwen3 is shown filled, Qwen3.5 as triangles, Gemma 3 open. In the positive control, where an indirect path is the known mechanism, route accounting finds it as the largest path, so the small purple shares elsewhere are not a blind spot of the method. Interactions are reported in the sections.

order, that the model rhymes, with donors of a different rhyme group drawn with seed 20260925. Hidden-choice lists are drawn from 8 single-token fruits (4-item lists) or from the first 12 single-token animals of a fixed 16-word pool (6-item lists), with seed 20260925. Chains and sums are drawn with the same seed, balanced over first digits.

Batching. Where batched, forward passes group sequences of equal length only; we found that left padding shifts positions in patched forward passes. Batched and unbatched cells agree to a median of 0.05 and a maximum of 0.13 log-odds in a reference check. Batched generation uses left padding with per-row patches. It reproduces unbatched greedy generations in 7 of 8 checks, and the remaining one is a bfloat16 near-tie.

C THE SIX-CELL ASSAY FOR SINGLE-TOKEN MEDIATORS

For a mediator token B with support $\{b_0, b_1\}$, the six cells are $I \in \{0,1\} \times B \in \{\text{free}, \text{do}(b_0), \text{do}(b_1)\}$. Free generation is a policy, not a third value of B. With B_i the greedy mediator under $I = i$ and $Y(i, b)$ the continuation distribution, the total effect $Y(1, B_1) - Y(0, B_0)$ splits exactly into emission $Y(0, B_1) - Y(0, B_0)$ and persistence $Y(1, B_1) - Y(0, B_1)$. Under a stochastic policy $q_i(b)$, emission is $\sum_b [q_1(b) - q_0(b)]\, Y(0, b)$, which gives the leverage identity of Appendix Section G for binary mediators. The multi-token version of Section 2 replaces the mediator with the generated text g.

D DECISION RULES COMMITTED BEFORE THE RESULTS

After the initial couplet split, every experiment's design, decision rule and predictions were committed to the project repository before its first result. These are commitments in version control, not entries in an external registry. Table A1 lists each rule with the commit that recorded it (committer time, UTC+9) and its outcome. Deviations are dated in each experiment's record. The main ones:

- the text-swap reference in the hidden choice, added after a 3-pair smoke test at 1.7B and before any real run;
- pick specificity, added post hoc for fruits, then committed as the primary measure of the animal replication before its result;
- the absolute-effect reading rule for distance (a rise in the indirect share counts only if the absolute indirect effect also grows), added after the full-attention control and before the Gemma 3 result;
- the fast path for the Qwen3.5-27B pilot (24 couplets, screen generations reused), committed before its result;
- the block edit for chains, added after the first 8B rows and before any 14B or 32B run;
- for the submission deadline, the 27B window curve shortened to 18 couplets, and the following committed checks not run, recorded before the affected results: the Gemma 4 path split and induction control, the Qwen3.5 mixture-of-experts model, the Qwen3.5-27B recurrent test at 100 couplets (the decision rests on 24), the Gemma 3 27B base model in the plain format, and the Qwen3-32B same-pick null.

E POSITIVE CONTROL: DETAILS

The design is 100 random sequences of 24 distinct common word tokens, repeated up to the second A. The edit replaces the first A's states with those of a sequence identical except for that token. The outcome is $\log p(B)$ at the second A. The control passes cleanly in four models and partly in five (Figure A1; Table A2).

- *Clean pass.* In Qwen3 1.7B, 4B, 8B and 14B, direct retrieval is 1 to 3% of persistence, and the indirect path is 108 to 137%. Shares above 100% reflect an interaction: with only the downstream positions edited, the clean first A conflicts with the edited "preceded by" signal and suppresses B more than the consistent edit does.
- *Partial pass, from the same interaction.* In Gemma 3 12B and 27B, direct retrieval is 0.4%, but the indirect path through all intermediate positions is only 53% and 55%. Editing B's first position alone gives 145% and 156%, so the conflicting edited positions weaken the all-position cell.

Table A1: Decision rules and predictions, the commit that recorded each before its result, and the outcome.

Experiment	Decision rule or prediction	Committed	Outcome
Relay grows with scale (couplets)	(1) the indirect share rises with log size; (2) the rise comes from relay (line-2 positions before the last three); (3) replicates in Gemma 3	d4622d2, 09-25 23:11	(1) holds, +6.2 pp/decade [2.8, 9.6]; (2) fails, so relay does not grow
Plain-format couplets	direct retrieval dominates; stored copy large in Gemma 3 27B, small in Qwen3-32B; the Qwen3 copy grows with scale	34c0cbd, 09-25 23:59	all three hold
Necessity split	report necessity with sufficiency; Shapley summary	f201b42, 09-26 02:45	reported for 17 runs
Induction control	indirect path $>80\%$ of persistence, direct retrieval near zero	01c27f1, 09-26 03:18	clean in Qwen3 1.7–14B; partial in Gemma 3 12B/27B (indirect 53–55%) and Qwen3-32B, Qwen3.5-9B, Qwen3.5-27B (direct retrieval 31%, 10%, 17%)
Sums	computation-position edit must move the answer (efficacy gate)	7f5a9a6, 09-25 22:05	gate fails at every size; downstream block $\leq 6.4\%$
Variable chains	indirect path through the chain $\geq 25\%$ of persistence (positive-control outcome)	59af2bc, 09-26 00:23	not observed: downstream block $\leq 4.9\%$
Written chains	emission accounts for nearly all of the effect	694af09, 09-26 05:03	holds (83–101%; 101% where the bypass is negative)
Recurrent memory	go if recurrent-only $\geq 25\%$, stop if $<10\%$ (largest model)	784c78b, 09-26 10:24	stop (2% at 27B)
Distance	go if the indirect path through the generated text $\geq 25\%$, stop if $<10\%$, else extend	2df09ce, 09-26 11:56; v2 6caf0e5; extension 0dda3b3	extended; 12%: between stop and go
Window curve (27B)	the change in reliance occurs between 1,000 and 1,500 tokens	a2d7c65, 09-26 15:43	partly supported: largest point-estimate step there, but the shift begins at 500 tokens (interior points 18 couplets, endpoints 50)
Window curve (12B)	plan survives inside the window, lost outside	9edfacc, 09-26 17:00	not supported: gradual fade from 500 tokens (distance, not the window)
Hidden choice	mostly not reproduced by the post-list patch (list large, post-list and sentence small) vs stored or passed through the sentence	144f545, 09-26 06:47	mostly not reproduced by the post-list patch; small stored part; through the sentence $\leq 2\%$ of the text swap
Out-of-list control	post-list edit raises the absent donor pick	aaa8b75, 09-26 07:23	holds in every model
Animal replication	pick specificity positive, growing in Gemma 3, near zero in Qwen3 at 14–32B	061b175, 09-26 09:34	Gemma 3 partly (4B $>$ 12B); Qwen3 small (3%, 7%)
Localization robustness	storage concentrates on one or two function words and moves with the wording	0ae75d6, 09-26 14:40	supported at 4B, 12B and 27B (the site moves with the rewording at all three)
Same-pick null	a same-pick donor leaves pick preference unchanged	a2d7c65, 09-26 15:43	holds at Gemma 3 12B (-0.19 $[-0.53, 0.16]$) and 27B (-0.42 $[-0.96, 0.10]$)
Architecture generalization	relay $<5\%$; induction direct retrieval $<5\%$; hidden choice through the sentence near zero; family-like storage	ead2545, 09-26 14:50	Qwen3.5-27B: storage 3–4%, stored part through the sentence 1% (as predicted), induction direct retrieval 17% (fails); Gemma 4 31B storage 82–91% (family-like, as predicted); other cells not run

Table A2: Positive control: induction. Effects on $\log p(B)$ of the edit at the first A (95% intervals), 100 sequences per model. *Key position*: only B's first occurrence receives its edited-run state. The criterion committed before the result: the indirect path above 80% of persistence and direct retrieval near zero.

Model	$p(B)$, clean	Persistence	Direct retrieval	Indirect (all positions)	Key position only
Qwen3-1.7B	1.000	-4.26 [-4.60, -3.91]	-0.08 [-0.13, -0.03]	-5.85 [-6.63, -5.10]	-9.24 [-10.21, -8.26]
Qwen3-4B	1.000	-4.75 [-5.06, -4.43]	-0.15 [-0.27, -0.06]	-5.23 [-6.10, -4.39]	-10.28 [-11.45, -9.08]
Qwen3-8B	0.999	-3.83 [-4.11, -3.54]	-0.05 [-0.09, -0.02]	-4.13 [-4.83, -3.45]	-10.20 [-11.40, -9.02]
Qwen3-14B	0.998	-3.81 [-4.14, -3.46]	-0.10 [-0.15, -0.05]	-4.64 [-5.33, -3.95]	-8.86 [-9.82, -7.88]
Qwen3-32B	0.977	-4.83 [-5.17, -4.48]	-1.52 [-1.87, -1.19]	-4.30 [-4.86, -3.75]	-6.24 [-6.93, -5.56]
Gemma 3 12B	0.999	-4.90 [-5.18, -4.64]	-0.02 [-0.03, -0.02]	-2.61 [-3.17, -2.11]	-7.12 [-8.16, -6.08]
Gemma 3 27B	0.998	-4.55 [-4.82, -4.27]	-0.02 [-0.02, -0.02]	-2.52 [-2.96, -2.10]	-7.10 [-8.05, -6.16]
Qwen3.5-9B	0.988	-3.81 [-4.01, -3.62]	-0.37 [-0.44, -0.31]	-2.33 [-2.67, -2.01]	-3.66 [-4.22, -3.12]
Qwen3.5-27B	0.984	-4.34 [-4.55, -4.12]	-0.72 [-0.85, -0.61]	-2.41 [-2.69, -2.15]	-3.35 [-3.90, -2.83]

- *Partial pass, because the model also reads the source.* The indirect path remains the main path in Qwen3-32B (89%) and in the recurrent hybrids Qwen3.5-9B and 27B (key position alone 96% and 77%). But direct retrieval is 31%, 10% and 17%: these models also read the first A directly.

In every model, the indirect path is the largest path, so the method detects an indirect path where one is the mechanism. The key position, B's first occurrence, lies right after the source in the prompt. Structurally this path therefore resembles a stored copy more than relay through generated text. What the control establishes is that route accounting detects an effect that arrives through an intermediate position. Relay is the same kind of path through different positions. Persistence holds the generated text fixed, so during measurement generated tokens are ordinary context positions, and the clamp is the same operation at any position. The lookup-dominated results in the rest of the paper are therefore informative negatives, not a blind spot of the method.

F RHYME PLANS: DETAILS

Task. Rhyming couplets are the flagship case of latent planning. The model commits to a rhyme at the end of line 1 and realizes it at the end of line 2 (Lindsey et al., 2025; Hanna & Ameisen, 2026a). We use the couplet prompts released by Hanna & Ameisen (2026a) and their chat format, with thinking disabled. For Qwen3 1.7B to 14B we use their 100-couplet steering subsets (Hanna & Ameisen, 2026b). For Qwen3-32B and Gemma 3 we apply their selection rule: the first 100 first lines the model rhymes, judged by their perfect-rhyme criterion. The source is their anchor, the last word of line 1. The donor is a couplet from a different rhyme group. The target is the position that predicts the first token of line 2's last word, with the line-2 prefix fixed. We repeat every run with the plain prompt of Ma & Rui (2026) (`A rhyming couplet:` followed by line 1 and a newline), with no chat template.

Behaviour. The all-layer edit of the rhyme word moves line 2 off its rhyme: after the edit, at most 2% of second lines rhyme with the original, against 83 to 100% before it. It moves line 2 onto the donor's rhyme in 27 to 75% of couplets in Qwen3 and 62 to 95% in Gemma 3. The emitted line-2 words account for 13 to 25% of the total effect. The rest is persistence: the rhyme position is pushed toward the donor even when the line-2 words are held fixed.

Growth with scale. Before running the larger models we committed two criteria for the claim that relay grows with scale. First, the share of persistence that takes an indirect path must rise with log parameter count across Qwen3 1.7B to 32B. Second, the rise must come from line-2 positions more than three tokens before the rhyme word, that is, from relay, because the published circuit already retrieves the plan into the last few positions (Hanna & Ameisen, 2026a). The first criterion holds: the per-couplet indirect share rises by 6.2 percentage points per decade of parameters (95% interval [2.8, 9.6]; medians 4, 8, 8, 11 and 13%). The second fails. Relay is -0.06, $+0.07$, $+0.04$, -0.01 and $+0.16$ at 1.7B, 4B, 8B, 14B and 32B, against persistence of 12 to 26. By the readings committed in advance, relay does not grow with scale. The indirect share grows as the late lookup up to 8B and, from 14B, as the stored copy at the end of line 1 (Section 3.2).

Controls. Three checks rule out that the split reflects disruption by the edit, the choice of source position, or the edit type.

Table A3: Couplet paths for every run (100 couplets; shares of persistence with the original line-2 words). Total and emission in log-odds; donor rhyme rate after the edit. *Retrieval*: direct retrieval of the rhyme word. *Indirect*: all intermediate positions together, split into *Late* (the late lookup), *Stored* (the stored copy) and *Relay*, which need not sum exactly to it. *Gap*: the interaction, persistence minus direct retrieval minus indirect.

Run	Donor rhyme	Total	Emission	Persistence	Retrieval	Indirect	Late	Stored	Relay	Gap
Qwen3-1.7B	27%	14.4	2.7	12.1	95%	7%	6%	0%	−1%	−1%
Qwen3-4B	43%	19.5	3.1	15.3	91%	10%	10%	0%	0%	−2%
Qwen3-8B	61%	26.0	3.6	22.6	90%	11%	10%	0%	0%	−1%
Qwen3-14B	64%	30.2	5.6	25.8	82%	13%	6%	5%	0%	5%
Qwen3-32B	66%	27.3	5.8	22.2	82%	16%	8%	8%	1%	2%
Qwen3-1.7B, plain	36%	10.1	2.4	9.0	93%	6%	5%	0%	0%	1%
Qwen3-4B, plain	54%	17.8	2.4	16.5	91%	11%	10%	1%	1%	−2%
Qwen3-8B, plain	69%	21.0	3.0	19.1	89%	10%	9%	1%	0%	0%
Qwen3-14B, plain	75%	23.8	3.9	19.1	76%	22%	9%	13%	0%	3%
Qwen3-32B, plain	70%	23.0	5.4	18.2	79%	19%	8%	10%	1%	2%
Qwen3-8B-Base, plain	68%	14.9	2.5	13.3	92%	10%	9%	1%	0%	−2%
Qwen3-14B-Base, plain	71%	18.2	4.0	14.6	77%	23%	7%	16%	0%	0%
Gemma 3 1B	62%	27.6	3.7	24.8	95%	4%	4%	0%	0%	1%
Gemma 3 4B	86%	50.3	7.3	41.4	95%	2%	1%	1%	0%	3%
Gemma 3 12B	95%	52.2	11.3	41.2	89%	2%	2%	0%	0%	8%
Gemma 3 27B	93%	51.8	7.0	43.1	68%	3%	1%	2%	0%	29%
Gemma 3 1B, plain	90%	31.5	6.9	24.9	95%	5%	4%	1%	0%	0%
Gemma 3 4B, plain	95%	43.3	8.7	34.4	94%	5%	4%	0%	0%	1%
Gemma 3 12B, plain	91%	39.1	9.8	27.9	85%	9%	6%	2%	1%	5%
Gemma 3 27B, plain	94%	36.4	9.2	26.9	19%	48%	7%	37%	1%	33%
Gemma 3 12B PT, plain	73%	16.7	4.0	13.7	80%	19%	13%	5%	1%	1%

- *Same-rhyme donors.* A donor from the same rhyme group leaves the rhyme almost unchanged. Persistence is $+0.1$ to $+2.0$, against $+12.1$ to $+22.6$ with different-rhyme donors in Qwen3 1.7B to 8B, and $+1.6$ against $+41.2$ in Gemma 3 12B. The next-word distribution still changes substantially, so the measure tracks rhyme identity, not generic disruption.
- *Anchor specificity.* Donor states at the rhyme word move the rhyme about two to three times more than donor states at any other single position of line 1 (the first word, a middle word, the punctuation, the end of the turn). Rhyme information at those positions is still partly retrieved directly, at a fifth to a half of the anchor's effect. Direct retrieval is weighted toward the rhyme word rather than confined to it.
- *Published feature steering.* We reselect the rhyme features of Hanna & Ameisen (2026a) with their rule. The mean is 19.1 features per couplet against their 19.2, with an exact per-couplet count match on 67 of 100 couplets at 4B. We then steer with their protocol instead of editing the state. Direct retrieval accounts for 90%, 88% and 84% of persistence at 4B, 8B and 14B, and indirect paths for 8%, 9% and 14%. The state edit gives the same split (91/10, 90/11 and 82/12), so its strength does not drive the result.

The Qwen3.5 hybrids show the same structure of paths (direct retrieval 84 to 88% of persistence; Section 4.2).

Table A3 gives the path split for every couplet run. Shapley values for the stored copy (the average of sufficiency and necessity) are $+1.73$ (Qwen3-14B), $+2.07$ (Qwen3-32B), $+2.78$ (Gemma 3 27B), $+2.81$ (Qwen3-14B plain), $+2.05$ (Qwen3-32B plain) and $+0.77$ (Gemma 3 12B plain).

Anchor specificity (persistence, donor states at one position of line 1). At Qwen3-1.7B: the rhyme word $+12.1$, end of turn $+6.0$, punctuation $+5.7$, a middle word $+5.5$, the first word $+3.5$. At 8B: $+22.6$, $+10.6$, $+10.7$, $+8.6$ and $+5.1$. At 32B: $+22.2$, $+11.0$, $+11.5$, $+4.5$ and $+3.8$. At Gemma 3 12B: $+41.2$, $+15.3$, $+14.2$, $+10.9$ and $+7.9$. The indirect path is small from every source position.

Converging evidence from the released attention intervention. Hanna & Ameisen (2026a) block the heads that read the rhyme word near the end of line 2. Afterwards only 30%, 20%, 13% and 32% of couplets still rhyme at 1.7B, 4B, 8B and 14B, all of which rhymed before. Direct retrieval is necessary, and no indirect path replaces it.

F.1 The stored copy

Almost all of the copy sits on the line's final token. In the five runs with a per-token analysis, the line's final token accounts for 90 to 100% of the boundary positions' total positive necessity

(Qwen3-14B 98%, Qwen3-32B 95% and Gemma 3 27B 93% with the chat prompt; Gemma 3 12B 90% and 27B 100% with the plain prompt). Necessity is measured one token at a time over the positions between the rhyme word and line 2. Every boundary token is reset to its clean state in turn, with the rhyme word edited (Figure A2 A).

- In Qwen3-14B and Qwen3-32B with the chat prompt, the stored copy sits entirely on the comma that ends line 1, with necessity $+2.09$ [1.50, 2.79] and $+2.02$ [1.46, 2.67]. Every chat-template token (<|im_end|>, the role header, the empty think block, the newlines) is 0.00 ± 0.03 by both measures.
- Gemma 3 27B shows the same site: the comma has necessity $+3.45$ [2.03, 5.05], and the other boundary tokens are 0.00 to 0.11, with intervals that include zero.
- With the plain prompt, where line 1 ends in a newline, the copy moves to the newline. In Gemma 3 12B the newline has necessity $+0.68$ [0.41, 0.98] and the comma $+0.07$. In Gemma 3 27B, where the copy is necessary for 74% of persistence, it sits entirely on the newline ($+19.7$ [18.1, 21.4]; the comma 0.00).

The storage site is therefore not a fixed token type. It is the last token of the line, whatever that token is. This is the position at which Lindsey et al. (2025) observed Claude's rhyme plan being formed.

The copy appears from about 12 to 14B parameters, and the base checkpoints we tested have it too. Table A4 gives the copy's share of persistence by both measures (bottom row of Figure 2).

- In Qwen3 the copy is absent up to 8B (necessity at most 0.9% of persistence in either format). It appears at 14B: 8% of persistence in the chat format and 16% in the plain format, by necessity. It remains at 32B (11% and 13%).
- In Gemma 3 it is absent at 1B and 4B, small at 12B (1 to 3%), and present at 27B: 11% by necessity in the chat format, and in the plain format 37% by sufficiency and 74% by necessity ($+19.8$ [18.2, 21.5] of $+26.9$). The plain format at 27B is the one configuration in which more of the plan is read from the line's final token than from the rhyme word (19%): a near-complete hand-off to the stored copy, while relay stays at 1%.
- The base checkpoints store it as well. Qwen3-14B-Base stores 16% by sufficiency, against 13% for the instruct model in the same format; Gemma 3 12B PT stores 5% and 6% against 2% and 3%. Qwen3-8B-Base, like the 8B instruct model, stores nothing. In the three base checkpoints tested, the copy is present wherever the matching instruct model has it, so it most likely arises in pretraining rather than from instruction tuning.
- With same-rhyme donors, the stored copy vanishes (-0.02 to $+0.01$ at 8B to 32B, against $+0.06$ to $+1.72$ with different-rhyme donors). It encodes the rhyme, not the disruption of the edit.

Where paths interact, the copy matters more than its sufficiency suggests. In Gemma 3 27B with the chat prompt, the stored copy alone moves the rhyme by only $+0.68$, yet resetting it removes $+4.87$ [3.18, 6.73] of a persistence of $+43.1$. The additivity gap is 29% of persistence (Qwen3 at most 5%). The target combines the rhyme word and the copy. When only one of them is edited, it reads conflicting evidence and moves less than the two effects would suggest. This is redundant storage: a second lookup site that backs up the first. Reporting sufficiency alone would miss most of it. Shapley values, the average of the two measures, are given in Appendix Section F.

Reconciling published accounts. Accounts of rhyme planning disagree on where the plan lives. The disagreement dissolves once both lookup sites are measured.

- Hanna & Ameisen (2026a) find that attention heads near the end of line 2 read the rhyme word, and that blocking them sharply reduces rhyming at every size up to 14B. In our terms this is direct retrieval together with the late lookup, which dominate at every Qwen3 size.
- Lindsey et al. (2025) find Claude's plan at the newline ending line 1, and Maar et al. (2026) steer successfully at the end of the line. This is the stored copy, which appears in larger models.
- Ma & Rui (2026) patch single layers in plain-text couplets. They report a hand-off from the rhyme word to the newline in Gemma 3 27B only, and none in Qwen3-32B. Our plain-format measurements agree: Gemma 3 27B reads more of the plan from the stored copy (37%) than from the rhyme word (19%). Qwen3-32B keeps a partial copy (10 to 13%), which a single-layer patch of the naturally computed state can plausibly miss.

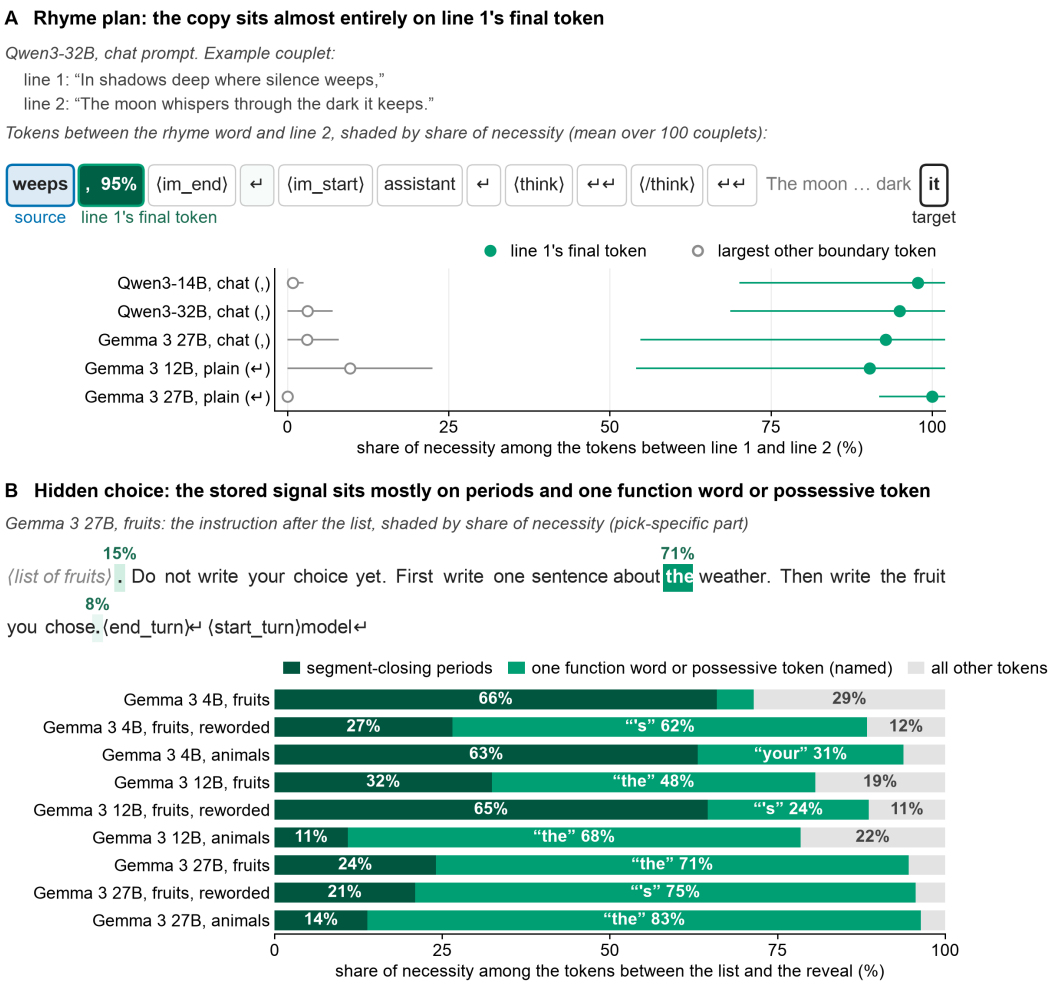


Figure A2: Where stored copies sit. Every value is a token's *share of necessity*: its interventional necessity (the loss of the effect when that token alone is reset to its clean state, with the source edited) divided by the summed positive necessity of all tokens examined, so the shares of a row add up to 100%. Cell shading and printed percentages show the same quantity. **A:** rhyme plan. Top, an annotated example (Qwen3-32B, chat prompt): of all tokens between the rhyme word and line 2, only the comma that ends line 1 matters. Below, for every model with a per-token analysis, the share on the line's final token (the comma before the chat template, or the newline in the plain prompt) against the largest share on any other boundary token. **B:** hidden choice. The instruction with its storage tokens highlighted, and for each model and wording the share of the pick-specific information on the segment-closing periods, on one function word or possessive token (*the*, *your*, or the apostrophe of "today's"), and on all other tokens. At 12B and 27B it sits mostly on *the* in "about the weather"; at 4B on the periods; with the reworded instruction it moves to the list-closing period and the token before "weather". The full token maps are in Appendix Figure A5.

- Jacopin (2026) finds no newline effect at 0.6B to 2.6B. We also find no copy below 8B.

Each account is correct for the configuration it studies. The proportions of the two sites depend on scale, family and prompt format.

G Computed and written values

Sums. The prompt poses a two-digit addition. The model writes an answer-neutral filler of 0 to 32 tokens and then the answer (104 items per model, Qwen3 1.7B to 32B; accuracy 95 to 100%).

Table A4: The copy at the line's final token, as a share of persistence (100 couplets per run). Sufficiency: the boundary positions alone receive their edited-run states. Necessity: the boundary positions alone are reset with the rhyme word edited. –: not run.

Model	Chat: sufficiency	Chat: necessity	Plain: sufficiency	Plain: necessity
Qwen3-1.7B	0%	−4%	0%	1%
Qwen3-4B	0%	−1%	1%	0%
Qwen3-8B	0%	0%	1%	1%
Qwen3-8B-Base	–	–	1%	0%
Qwen3-14B	5%	8%	13%	16%
Qwen3-14B-Base	–	–	16%	–
Qwen3-32B	8%	11%	10%	13%
Gemma 3 1B	0%	0%	1%	0%
Gemma 3 4B	1%	–	0%	–
Gemma 3 12B	0%	1%	2%	3%
Gemma 3 12B PT	–	–	5%	6%
Gemma 3 27B	2%	11%	37%	**74%**

The committed design named the question mark as the position where the sum would be computed. Donor states there move the answer by -0.08 to $+0.07$ log-odds, so the efficacy gate fails at every size. That shows only that the question mark does not hold the sum. The stronger test edits every position from the question mark to the answer together: this block moves the answer by at most 6.4% of the effect of donor operand digits ($+17$ to $+54$; 6.4% at 32B, $+1.10$ [0.93, 1.27] of $+17.2$, and at most 3.4% at the smaller sizes). The downstream block is small beside the operand edit, which is consistent with the answer re-reading the visible operands; recomputation itself was not measured.

Variable chains. Sums are cheap to recompute. A chain of updates (`a = 23, b = a + 4, c = b + 7, ..., What is e?`) makes recomputation at the answer costly and makes an indirect path the natural one. The running value could be passed from statement to statement, so this task was designed as a second positive-control candidate for an indirect path. It is not one. Donor states at the starting value's digits move the answer strongly (at 14B: $+30.9$, $+18.5$ and $+6.4$ for $K = 1, 3, 5$ updates). Where single positions were edited (1.7B to 14B; at 32B only the block was run), donor states at any later position, or at any statement end, move it by less than 0.4 in absolute value. Donor states at *every* position after the starting value together, a test that would detect a running value stored redundantly, move it by at most 4.9% of the source-digit effect (32B, $K = 1$: $+0.81$ [0.61, 1.03] of $+16.5$). The other cell whose interval excludes zero is 14B at $K = 3$ ($+0.44$ [0.08, 0.82] of $+18.5$, 2.4%). At 32B the block is -0.02 [-0.33, 0.29] of $+10.1$ at $K = 3$ and -0.00 [-0.22, 0.19] of $+5.2$ at $K = 5$. In the remaining size and length combinations the interval includes zero. Figure A3 shows the $K = 3$ series, which stays near zero. The downstream block is small beside the source edit, which is consistent with the answer re-reading the chain. Accuracy is consistent with recomputation at the answer. It falls steeply with chain length, from 97 to 100% at $K = 1$ to 1 to 17% at $K = 5$ (Appendix Figure A4). At $K = 5$, where maintaining a running value would make more answers correct, the downstream block's interval includes zero at every size. That agreement is correlational, not a test of what the model could do.

In these chains the answer follows the written values. We return to the variable chains of Section 3.3, now with the instruction "Compute each variable in order, one per line, then state the value of e". The model writes each intermediate value. The edit at the starting value's digits makes it write the donor's values in 86 to 100% of items. We cut the written chain after the second-to-last value and measure the answer's first digit. The six-cell split then shows how much of the effect goes through the written values (emission) and how much bypasses them (persistence with the text fixed).

- Emission accounts for 92% and 96% of the effect at 1.7B ($K = 3$ and $K = 5$), 93% and 96% at 4B ($K = 3$ and $K = 5$), 94% and 92% at 8B, 88% and 83% at 14B, and 98% and 101% at 32B (every chain usable after the parser fix).
- The bypass, which re-reads the edited source digits, is 4 to 8% at 4B and 8B and 12 to 17% at 14B. At 32B it is $+0.15$ [0.03, 0.28] (2%) at $K = 3$ and -0.08 [-0.17, 0.01] at $K = 5$. At 32B and $K = 5$, emission is 101% of the total because that bypass is negative. Emission plus the bypass equals the total ($+6.62$ and -0.08 sum to $+6.54$).

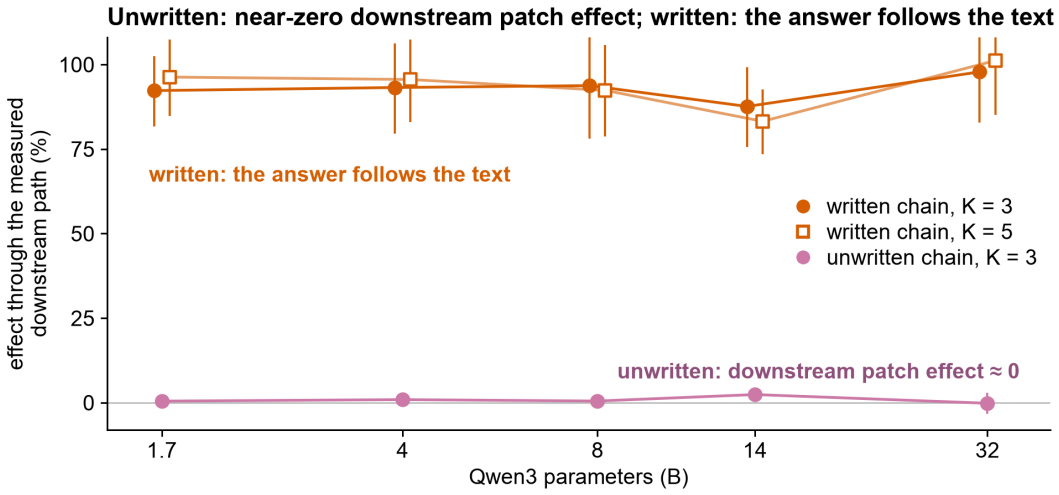


Figure A3: Unwritten versus written computed values (Qwen3 variable chains), on one axis with two denominators. *Unwritten:* the effect of donor states at every position after the source, as a share of the effect of editing the source digits. *Written:* emission divided by the total effect. 95% intervals. For unwritten chains ($K = 3$ updates) this downstream patch effect stays near zero, which is consistent with the answer re-reading the source; recomputation itself is not measured. *Written*: the share that goes through the values the model writes (emission; $K = 3$ and $K = 5$); the answer follows the text. The remainder of the written case is the bypass to the edited source. The two series cover different sizes because of which runs exist: both series now cover 1.7B to 32B; at 32B the unwritten point comes from the block edit alone.

Written reasoning is not decorative in these models. The answer follows the written values, and the part that re-reads the source stays below 17% at every size tested. The couplets give the same picture on a smaller scale: the emitted line-2 words account for 13 to 25% of the rhyme effect (Section 3.1).

Leverage. How much of an effect goes through a written token depends on how much that token matters. For a binary mediator $B \in \{b_0, b_1\}$, such as the article *a* or *an* before a planned noun, let $q_i = P(B = b_1 \mid I = i)$ be the probability of writing b_1 with the intervention off ($i = 0$) or on ($i = 1$). Let $Y(0, b)$ be a scalar outcome, here $\log p$(planned noun), with the intervention off and the mediator fixed to b. Policy-weighted emission is defined as $E_{\text{emit}} = \sum_b [q_1(b) - q_0(b)]\, Y(0, b)$ (Appendix Section C). For a binary mediator it equals, by definition,

$$E_{\text{emit}} \;=\; (q_1 - q_0)\,\big(Y(0, b_1) - Y(0, b_0)\big).$$

This is the change in the policy-weighted expected outcome, not the change in the logarithm of the marginal probability. The effect through the text is the change in what is written, times how much the written token matters to what follows. We call the second factor the token's *leverage*. We apply the six-cell assay to the a/an planning features of Hanna & Ameisen (2026a), which are selected by a rule with no article criterion. We use their per-layer transcoders for Qwen3 0.6B, 1.7B and 4B and reproduce their node counts exactly on 297 to 344 of 349 prompts.

- At 4B, where they report planning, the features have a bidirectional, feature-specific effect on the planned noun. Of that effect, 92 to 94% travels through the generated article. The effect that bypasses the article ($+0.012$ [0.000, 0.024] for amplified features) is at the level of randomly chosen features (-0.003 [-0.019, 0.013]).
- Stratified by leverage (Appendix Figure A4), the effect through the article scales with leverage, while the effect bypassing it stays small and constant. The article accounts for 53 to 82% of the effect in the lowest-leverage tertile and 94 to 96% elsewhere.
- The product of policy movement and leverage predicts the effect through the article better than policy movement alone (correlation 0.886 against 0.831 for amplified features).
- At 0.6B and 1.7B the effects are small and travel almost entirely through the article.

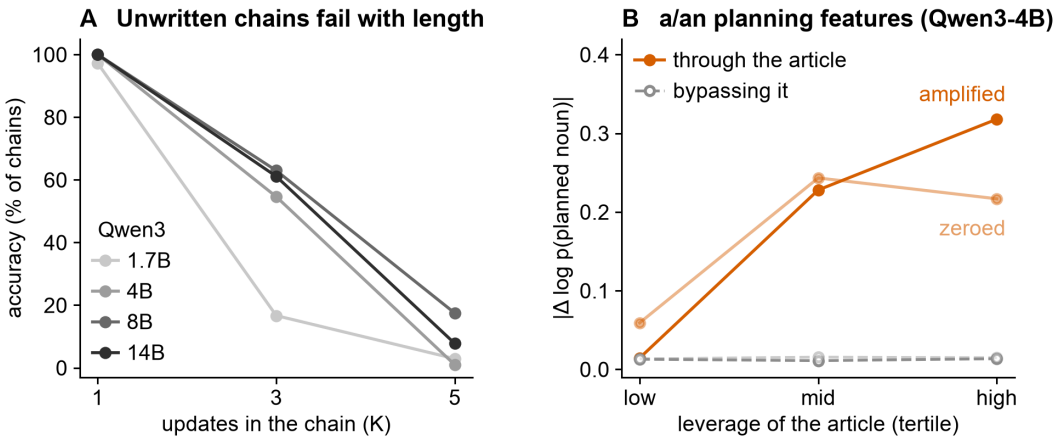


Figure A4: Computed values, supporting panels. **Left:** accuracy of unwritten variable chains (Qwen3 1.7B to 14B) against the number of updates. **Right:** a/an planning features in Qwen3-4B. The effect on the planned noun through the article (vermillion) against the effect bypassing it (grey), by tertile of the article's leverage over the planned noun, for amplified and zeroed features.

Table A5: Sums: donor states at each position group (log-odds toward the donor's first digit; 104 items).

	1.7B	4B	8B	14B	32B
Accuracy (all filler lengths)	95–100%	100%	100%	100%	100%
Donor state at `?` (committed)	+0.07	+0.01	−0.08	−0.06	−0.03
Every position from `?` to the target	+0.30	+0.58	−0.12	+1.62	+1.10
Operand digits	+47.8	+48.0	+53.6	+47.3	+17.2

Sum prompts. The user turn is `/no_think What is {a} + {b}?`. The assistant turn is prefilled with the filler and `The answer is`. The target is the first digit of the answer. Table A5 gives the efficacy scan.

Chain prompts. The user turn is `a = {v0}\nb = a + {d1}\n...\nWhat is {last}?`, with $K = 1, 3, 5$ updates, 103 to 108 items per K, and the same increments for original and donor, so the tokens align. Table A6 gives the edits.

Table A6: Variable chains (unwritten). Donor states at the starting value's digits versus at every later position (the block edit), $K = 1/3/5$.

	Accuracy	Source digits	Every later position
Qwen3-1.7B	97 / 17 / 3%	+36.3 / +24.4 / +13.6	−0.02 / +0.11 / +0.04
Qwen3-4B	100 / 55 / 1%	+47.1 / +32.2 / +20.1	−0.07 / +0.29 / −0.20
Qwen3-8B	100 / 63 / 17%	+45.1 / +31.1 / +18.4	+0.05 / +0.15 / −0.07
Qwen3-14B	100 / 61 / 8%	+30.9 / +18.5 / +6.4	+0.09 / +0.44 / +0.22

H PER-TOKEN STORAGE MAPS

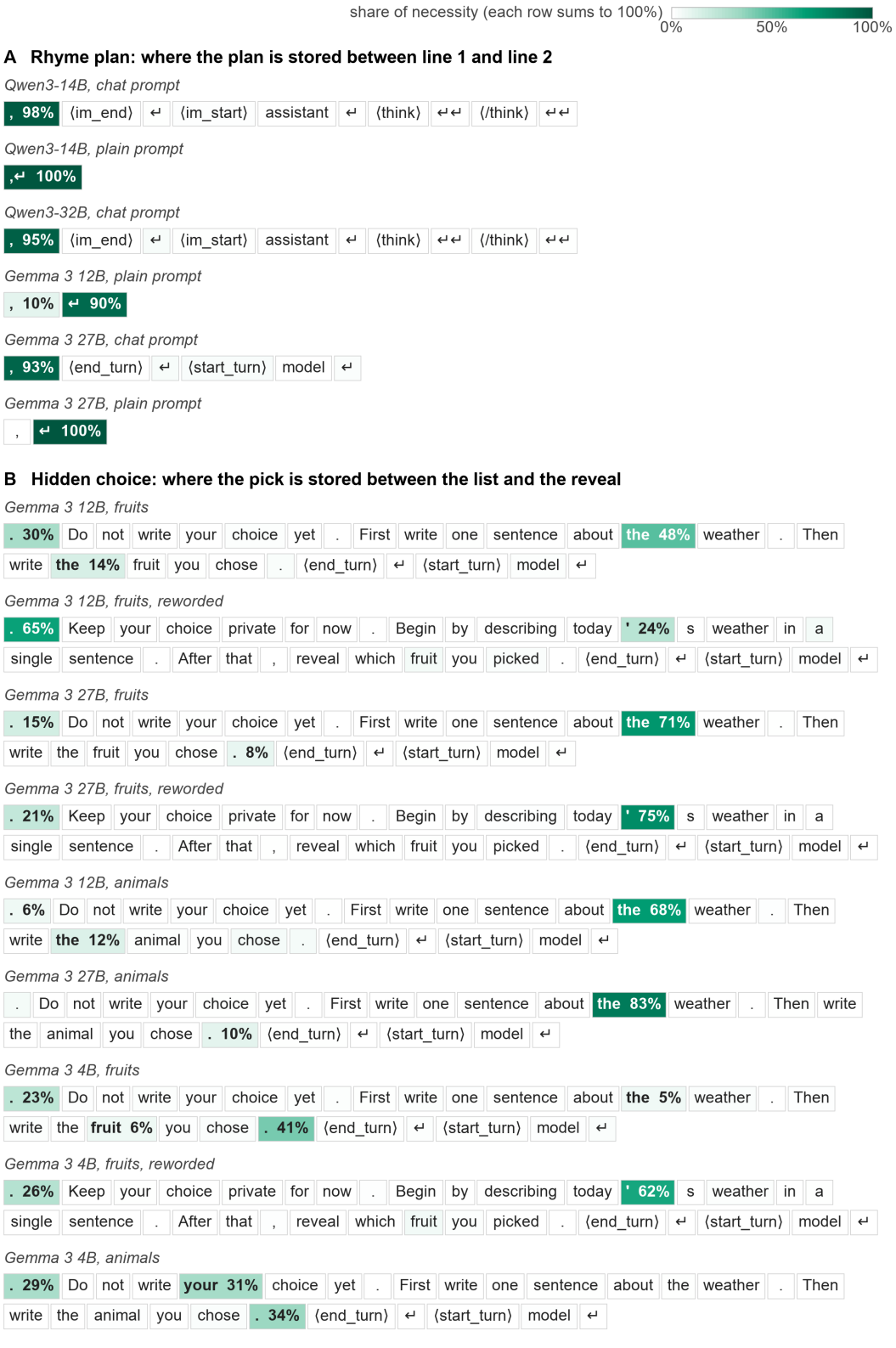


Figure A5: Per-token necessity for every model and wording analysed. Each cell is one token; the shade and the printed value give its share of necessity (Figure A2): the loss of the effect when that token alone is reset to its clean state, with the source edited, as a share of the row's summed positive necessity. Top rows: tokens between the rhyme word and line 2. Bottom rows: tokens between the hidden-choice list and the reveal (pick-specific part).

Table A7: Indirect paths at a distance, Gemma 3 27B (chat prompt; 50 couplets per distance; 95% intervals). Shares are of persistence at the same distance. The generated-text block is the cue plus all of line 2; it is not split into late lookup and relay.

	$D = 0$	$D = 2{,}000$
line 2 rhymes with line 1	98%	82%
persistence	+42.4 [38.6, 46.0]	+26.4 [23.8, 28.9]
direct retrieval	+28.0 (66%)	+10.5 (40%)
stored copy: sufficiency / necessity	+0.1 / +4.6 (11%)	+9.7 / +13.0 (49%)
filler: sufficiency / necessity	0.0 / 0.0	+0.3 / +0.15 [−0.26, 0.57]
generated text: sufficiency / necessity	+0.5 / +7.3	+3.2 (12%) / +2.7 (10%)

I SETTINGS BUILT TO FAVOUR RELAY: DETAILS

I.1 BEYOND THE ATTENTION WINDOW

Gemma 3 alternates five local layers with a 1,024-token sliding window (Beltagy et al., 2020) with one global layer (Gemma Team, 2025). Once line 1 lies more than 1,024 tokens before the rhyme position, five of six layers can no longer read it. We insert D tokens of neutral filler, a seeded arrangement of fixed everyday sentences, at the start of the model's turn, followed by a fixed cue (`Next line:`). The model then writes line 2. The indirect path is split into three groups of intermediate positions: the stored copy (the prompt positions after line 1), the filler, and the generated text (the cue and line 2). The committed decision number is the share of persistence that takes the indirect path through the generated text: "go" at 25% or more, "stop" below 10%, and extension to 50 couplets in between. A full-attention control, Qwen3-14B, showed before the Gemma result that indirect *shares* can rise with distance only because direct retrieval weakens. We therefore added, before that result, the rule that a rise counts only if the absolute indirect effect also grows.

Table A7 gives the extended run. At 2,000 tokens, Gemma 3 27B still writes a rhyming line 2 in 82% of couplets, but persistence falls from +42.4 to +26.4. The model reads less of the plan from the rhyme word: direct retrieval falls from 66% to 40% of persistence. It relies instead on the stored copy at the line's final token, which is necessary for +13.0 [10.0, 16.1], about half of persistence. The absolute indirect effect grows about thirteenfold (+1.0 to +13.1), mostly through the stored copy. The filler passes on nothing (+0.3 by sufficiency, +0.15 by necessity). The indirect path through the generated text rises from about 1% to 12% by sufficiency (+3.2 [2.2, 4.3]); its necessity is +2.7, or 10%. This block contains the cue and all of line 2, including its last three positions, so it mixes relay with the late lookup; the split was not run. It is the largest generated-text block we measure, a borderline case rather than a demonstrated exception: by the committed rule it falls between "stop" and "go", and how much of it is relay remains unresolved. Even at 2,000 tokens it is a minority path, about an eighth of the effect.

The window curve, committed before its result, predicted a change specifically between 1,000 and 1,500 tokens. The interior distances use 18 couplets each and the endpoints the 50 couplets of Table A7, so the curve is not one uniform sample. It is only partly supported. Direct retrieval is 66%, 64%, 54%, 40% and 40% of persistence at 0, 500, 1,000, 1,500 and 2,000 tokens, and the stored copy is necessary for 11%, 25%, 30%, 45% and 49%. The largest point-estimate step is where line 1 leaves the window, but the shift has begun at 500 tokens, inside it. The indirect path through the generated text is 11%, 9%, 13% and 12% by sufficiency at the four nonzero distances (by necessity 10%, 14%, 11% and 10%): it appears once filler is present and does not grow with distance. We therefore describe a long-distance shift toward the stored copy, steepest across the window, rather than a precisely located window-edge transition.

Three further models show what happens when lookup gets harder without a strong stored copy. They also separate distance from the window.

- *Gemma 3 12B*, which keeps only a small stored copy (Table A4), loses the plan. The window curve, committed before its result, predicted that the plan would survive while line 1 is inside the window (500 and 1,000 tokens) and vanish outside it. Instead, the plan fades gradually and is mostly gone inside the window. Second lines rhyme in 100%, 39%, 8%, 0% and 0% of couplets at 0, 500, 1,000, 1,500 and 2,000 tokens, and persistence falls from

+40.5 to +11.5 [8.1, 14.9] at 500 tokens and +4.0 at 1,000. At 12B the loss reflects the distance itself, not the window. No indirect path takes over: the absolute indirect effect stays at +0.8 to +2.3, and relay at or below +0.9.

- *Qwen3.5-9B*, a hybrid whose attention layers have no window, also fades: rhyming falls from 92% to 4% at 2,000 tokens. What remains of persistence (+3.6) is still 81% direct retrieval, and the absolute indirect effect does not grow (+1.2 to +0.8).
- *Qwen3-14B*, with full attention, largely abandons the task after the filler (8% rhyming), and its absolute indirect effect stays flat (+2.3 to +2.8).

Across the four models, distance moves reliance onto a stored copy where one exists, and otherwise costs the plan. In no model does an indirect path through the generated text take over. The 12% in Gemma 3 27B is the unsplit generated-text block, including its last three positions. The rhyme word and the line's final token lie at the same distance from the target, yet the global layers of Gemma 3 27B read the line's final token. This fits the view of separator tokens (Chen et al., 2025a) and label words (Wang et al., 2023b) as summaries of their segment.

I.2 Recurrent memory

Qwen3.5 (Qwen Team, 2026) interleaves three Gated DeltaNet layers (Yang et al., 2025b) with each full-attention layer. A recurrent layer holds a fixed-size state that is updated position by position. In those layers, information from the source can reach a later position only through that state, passed along every intervening position by construction and invisible in the text. The attention layers can still look it up. The test therefore asks which path the model uses when both are available. We split persistence by layer type. In the *attention-only* cell, the donor state enters at the source, but the input of every recurrent layer at the source is restored to its clean value, so the edit can move forward only through the attention layers. The *recurrent-only* cell is the converse. A cell with both layer types blocked is a sanity check. The rule committed in advance was "go" if recurrent-only is at least 25% of persistence with its interval above zero, and "stop" if it is below 10%. It is decided on the largest model; smaller sizes give the trend.

When the donor signal can enter at the source through either layer type, entry through the attention layers accounts for nearly all of its effect. At 4B and 9B (40 couplets each), the recurrent-only cell is +0.28 [0.07, 0.49] and +0.55 [0.31, 0.79]: 3% and 4% of persistence. The attention-only cell accounts for 97% and 96%. At the decisive size, 27B (48 recurrent and 16 attention layers; 24 couplets), recurrent-only is +0.41 [0.18, 0.64], 2% of persistence +18.8 [16.3, 21.3], and attention-only 95%. As a sanity check, blocking both layer types at the source cancels the donor: that cell is zero on every couplet, as it must be. Decision: stop. Although three quarters of the layers can pass the plan forward only through the recurrent state, a donor signal that enters through them at the source produces only 2 to 4% of persistence. The design controls which layer type receives the donor signal at the source; it does not isolate every later path through each layer type, so it measures the share of donor entry, not the recurrent memory's natural share. Entry through the recurrent layers is small but nonzero, which bounds the principle at 2 to 4% of persistence for this architecture.

J Hidden choice

In every task so far, the content is visible: a reader of the text could recover the rhyme sound from line 1, or the value from the operands. The case monitoring cares about is content that cannot be recovered from the text. We create it with a secret choice. The prompt reads: "Secretly choose one fruit from this list: {4 fruits}. Do not write your choice yet. First write one sentence about the weather. Then write the fruit you chose." Under greedy decoding the pick is a deterministic function of the prompt. It depends on the model's own preferences and on the list order, so no reader can compute it. For the path split, the model's turn is prefilled with a fixed weather sentence and "The fruit I chose is". The outcome is the log-odds of the donor's pick against the original pick at the next token.

A small part of the pick-specific signal is stored after the list, and almost none passes through the sentence. The donor is the same list in another order, chosen so that the greedy pick differs. The reference is a *text swap*: the donor order written into the prompt, which moves the pick by construction. In Qwen3-8B, Gemma 3 12B and Gemma 3 27B (100 pairs each), donor states at the prompt positions after the list move the reveal by 12%, 15% and 16% of the text swap. The reveal reads all of it directly: the direct-retrieval cell equals the post-list effect. The part of this stored signal that passes through the sentence is +0.00 [−0.04, 0.05], −0.06 [−0.14, 0.01] and −0.01 [−0.10,

Table A8: Stored pick-specific information: pick specificity of the post-list edit as a share of the text-swap pick specificity (100 pairs per cell; values with 95% intervals in Appendix Section J). Fruits: specificity added post hoc. Animals: committed before the result.

Model	Fruits (post hoc)	Animals (committed before the result)
Qwen3-8B	10%	15%
Qwen3-14B	0%	3%
Qwen3-32B	2%	7%
Qwen3.5-27B	3%	4%
Gemma 3 4B	18%	22%
Gemma 3 12B	15%	18%
Gemma 3 27B	31%	40%
Gemma 4 31B	82%	91%

0.08]; it is also zero within its interval in Qwen3-4B and Qwen3-14B. The rest of the pick, 84 to 88%, is not reproduced by the post-list patch; it is presumably re-derived at the reveal from the visible list. The two largest Qwen models are the exception.

- In *Qwen3-32B*, the indirect path through the sentence is small but nonzero: $+0.06$ [0.04, 0.09]. That is 2% of the text swap but one eighth of the stored part.
- In the recurrent hybrid *Qwen3.5-27B*, the part of the stored signal that passes through the sentence is $+0.03$ [0.02, 0.04]: 1% of the text swap and one sixth of the stored part. Its sentence positions themselves account for 7% of the text swap ($+0.15$ [0.12, 0.18]); Qwen3.5-9B gives the same pattern (through the sentence 2%, sentence positions 8%). These are the largest in any model; Qwen3-4B has 5% and every other model at most 1%. This is expected when every position's recurrent state summarizes the prefix, including the list.

What the stored signal could be. An order swap cannot separate a stored pick from a stored copy of the list order. In the out-of-list control, the donor list shares no item with the original, so the donor's pick does not occur in the prompt at all. The committed criterion is that post-list donor states raise the absent pick, and it holds in every model. After the first results we identified a confound: the positions might hold a copy of the donor's list rather than its pick. We therefore added, post hoc, *pick specificity*: the effect on the donor's pick minus the mean effect on the donor's other items. We then committed it, before the results, as the primary measure of a replication in a second domain (animals; disjoint lists of six). The replication confirms that pick-specific information is stored after the list, in both families (Figure 3 C; Table A8). A same-pick null, committed before its result, gives the converse check. The donor is the same list in another order with the *same* pick. Post-list donor states then leave the relative preference for the pick unchanged on average in Gemma 3 12B: -0.19 [-0.53, 0.16], against $+1.01$ [0.78, 1.24] with out-of-list donors. Gemma 3 27B gives the same null: -0.42 [-0.96, 0.10], against $+2.48$ with out-of-list donors.

These controls do not settle what the stored information is. The post-list positions could hold the pick itself. They could also hold a copy of the list from which the reveal re-derives the pick by applying the model's preference. Both accounts predict pick specificity with out-of-list donors, and both predict the same-pick null. We therefore claim only that information specific to the pick, and absent from the visible text, is stored after the list and read at the reveal.

The families diverge with scale.

- In Gemma the stored share grows with generation and size. Gemma 3 stores 18 to 22% at 4B, 15 to 18% at 12B and 31 to 40% at 27B. The committed monotone prediction (4B below 12B below 27B) is only partly supported, because 4B stores more than 12B. Gemma 4 31B stores 82% (fruits; $+2.93$ [2.56, 3.30]) and 91% (animals; $+4.44$ [3.97, 4.93]) of the text swap's pick specificity. For Gemma 4 the path split was not run.
- In Qwen3 the stored share shrinks with scale: 10 to 15% at 8B, 0 to 7% at 14B and 32B. Qwen3.5-27B stores 3 to 4%. In larger Qwen models the pick is mostly not reproduced by the post-list patch.

The written sentence has no measurable effect on the reveal. In the free version, the model writes its own weather sentence. The sentence changes with the donor in 17 to 45% of pairs, but it never names an item. Emission is $+0.02$ [-0.03, 0.08], -0.01 [-0.07, 0.04] and -0.00 in Qwen3-8B, Gemma 3 12B and Gemma 3 27B. The written sentence has no measurable causal effect on the reveal.

Where the stored signal sits. Per-token necessity over the post-list block localizes the stored information (Figure A2 B).

- In Gemma 3 27B, 77% of the pick-specific effect ($+1.91$ [1.57, 2.24] of $+2.48$) sits on a single token: *the* in "about the weather". With animals the same token accounts for 83% ($+3.36$ [2.92, 3.80] of $+4.07$). The period after the list accounts for $+0.42$, and the period ending the instruction $+0.22$. Every other token, including the end-of-turn and role tokens, is 0.00 ± 0.05.
- Gemma 3 12B uses the same site with fruits (*the*: $+0.52$ of $+1.03$; the period after the list: $+0.32$) and with animals (*the*: $+1.04$ [0.80, 1.28] of $+1.42$, 73%).
- Gemma 3 4B stores as much (18 to 22%) but places it mostly on the periods that close the list and the instruction (91% with fruits, 65% with animals), with a function word taking the rest (7% on *the* with fruits; 32% on *your* with animals).

The decisive check, committed before its result, changes the wording. If the storage moves with the function words, it is storage at a predictable low-content token. If it stays at a fixed position or spreads out, the site is template-specific. We reworded the instruction to "Keep your choice private for now. Begin by describing today's weather in a single sentence. After that, reveal which fruit you picked." Gemma 3 12B then stores 12% of the text swap's pick specificity ($+0.59$ [0.38, 0.81]), again on two low-content tokens. The period that ends the list accounts for $+0.42$ [0.21, 0.63] (71%), and the apostrophe of "today's", the token before "weather", for $+0.16$ [0.09, 0.22] (27%). Every other token is below 0.04. Gemma 3 4B moves the same way: the apostrophe of "today's" takes 69% ($+0.54$ [0.46, 0.62]) and the list-closing period 28%. Gemma 3 27B, where the base wording gave 77% on *the*, moves the same way. Its stored pick specificity is $+1.63$ [1.38, 1.88] (24% of the text swap's); of that, the apostrophe of "today's" accounts for 82% ($+1.34$ [1.14, 1.53]) and the list-closing period for 23% ($+0.37$), which Figure A2 B shows as 75% and 21% of the row's positive necessity. At all three sizes the site moved with the wording, from a function word to the possessive token before "weather".

Across sizes and wordings, the stored information sits on two or three low-content tokens. These are the punctuation that closes a segment (the list or the instruction) and one function word or possessive token of the instruction, whose identity depends on the wording (*the*, *your*, or the possessive before "weather"). Segment-closing punctuation is the analogue of the line's final token for rhymes. In both tasks the model stores a copy once, at a boundary next to the source or at a predictable token of the prompt, and reads it at the point of use.

At Qwen3-4B, the forced sentence positions hold a small part of the pick, $+0.57$ [0.48, 0.67] or about 5% of the text swap. The donor run's sentence positions read this directly from the list. In Qwen3 at 8B and larger and in Gemma 3 the sentence positions hold about nothing (Qwen3-32B about 1%); in the Qwen3.5 hybrids they hold 8% (9B) and 7% (27B, $+0.15$ [0.12, 0.18]) of the text-swap effect.

K Compute

Experiments ran on one Apple M1 laptop (16 GB) and one Apple M4 Max workstation (128 GB unified memory), with no GPU cluster. Jobs were scheduled under a declared memory budget and checkpointed per item, so interrupted runs resume without loss. A 27B to 32B couplet chain (screen, edit and generation, path split, position split and necessity) takes one to three hours. The hidden-choice runs take 10 to 30 minutes per model.

Table A9: Hidden choice, out-of-list control and replication (log-odds, 95% intervals). Specificity is the effect on the donor's pick minus the mean effect on the donor's other items.

Model	Domain	Text-swap specificity	Post-list: donor pick	Post-list: pick specificity
Qwen3-8B	fruits	+5.4	+2.01 [1.85, 2.18]	+0.55 [0.46, 0.64]
Qwen3-14B	fruits	+3.3	+0.29	−0.01 [−0.05, 0.03]
Qwen3-32B	fruits	+1.8	+0.20	+0.04 [0.01, 0.06]
Qwen3.5-27B	fruits	+1.66	+0.16 [0.11, 0.22]	+0.05 [0.03, 0.08]
Gemma 3 4B	fruits	+7.2	+4.85 [4.35, 5.35]	+1.33 [1.04, 1.61]
Gemma 3 12B	fruits	+6.9	+5.07 [4.65, 5.50]	+1.01 [0.78, 1.24]
Gemma 3 27B	fruits	+8.0	+10.30 [9.70, 10.90]	+2.48 [2.05, 2.91]
Qwen3-8B	animals	+4.30	+2.68 [2.46, 2.91]	+0.63 [0.50, 0.76]
Qwen3-14B	animals	+4.80	+0.80 [0.69, 0.93]	+0.13 [0.06, 0.21]
Qwen3-32B	animals	+1.64	+0.52 [0.40, 0.65]	+0.12 [0.05, 0.19]
Qwen3.5-27B	animals	+1.27	+0.18 [0.12, 0.23]	+0.05 [0.01, 0.08]
Gemma 3 4B	animals	+11.93	+8.17 [7.25, 9.09]	+2.59 [2.13, 3.09]
Gemma 3 12B	animals	+8.10	+4.94 [4.30, 5.59]	+1.43 [1.10, 1.77]
Gemma 3 27B	animals	+10.12	+14.24 [13.44, 15.05]	+4.09 [3.59, 4.58]
Gemma 4 31B	fruits	+3.59	+9.19 [8.88, 9.50]	+2.93 [2.56, 3.30]
Gemma 4 31B	animals	+4.90	+11.06 [10.67, 11.45]	+4.44 [3.97, 4.93]